Ground Moving Target Detection in Multichannel Polarization SAR-GMTI Systems

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***Abstract*—In this article, a novel ground moving target (GMT) detection algorithm is proposed in a synthetic aperture radar (SAR) system based on the space–time-polarization cascaded processing. In the proposed method, the polarization classification is first accomplished based on multipolarization data, followed by spatial domain adaptive processing in the SAR image domain for each polarization channel based on the polarization classification results. Subsequently, the preliminary detection results of moving targets from a single polarization channel can be obtained, and the target polarization steering vectors can be estimated through an iterative process, in order to achieve adaptive coherent synthesis of the target signal in the polarization domain and accomplish the secondary clutter cancellation. Furthermore, a target detection technique based on the polarization distance window detector (PD-DT) is proposed to effectively enhance the detection probability of moving targets and decrease false alarms. Finally, based on the estimated polarization steering vectors of GMTs, adaptive processing in the polarization domain is performed for each spatial channel, greatly improving the output signal-to-clutter-plus-noise ratios (SCNRs) for detected targets, beneficial to the subsequent target relocation. The effectiveness of the proposed algorithm is verified using real-measured data from an airborne multichannel fully polarimetric SAR (Pol-SAR) system.**

***Index Terms*—Adaptive processing, ground moving target indication (GMTI), multipolarization and multichannel, synthetic aperture radar (SAR).**

NOMENCLATURE

H/V	Horizontal (**H**) and vertical (**V**) polarization pulses.
PRT	Pulse repetition interval between two consecutive transmissions of the same polarization.
V_p	Platform speed.
H	Platform height.
D_a	Azimuth antenna length.
d_n	Equivalent phase centers of the transmitter and receiver.
$\Delta X/\Delta Y/\Delta Z$	Deviations of the aircraft from the ideal flight path along the range, azimuth, and altitude directions.
v_x/v_y	Range and azimuth motion velocities of the GMT.
t_m	Azimuth slow time.
v_r	Radial velocity of the GMT.
R_0	Target initial slant range.
t_r	Fast time.
t_p	Pulse duration.
T_a	Synthetic aperture time.
K_r	Fast-time chirp rate.
σ_{MT}	Scattering amplitude of the moving target.
$\theta_{Azi,c}$	Azimuth cone angle of the clutter scattering point relative to the radar receiver.
$\theta_{El,c}$	Depression cone angle of the clutter scattering point relative to the radar receiver.
$\theta_{cone,c}$	Spatial cone angle of the clutter scattering point relative to the radar receiver.
σ_c	Clutter scattering amplitude.
λ	Transmitted signal wavelength.
N	Number of receiving channels.
B_a	Signal azimuth bandwidth.
t_0	Center time of the aperture.
f_a	Doppler frequency.
R_{cn}	Clutter covariance matrix (CCM).

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I. INTRODUCTION

SYNTHETIC aperture radar (SAR) can effectively capture the scattering information of large ground scenes under all-weather conditions. Ground moving target indication (GMTI) technology can effectively monitor, track, and identify ground moving targets (GMTs) of interest [1]. Combining SAR technology with GMTI not only yields the high-resolution static scene images, but also accomplishes the effective detection, parameter estimation, and localization of GMTs within the observation scene [2], [3], [4], [5]. Owing to these capabilities, SAR-GMTI systems have found widespread applications in various civilian fields, including large-scale remote sensing, environmental and ecological monitoring,

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transportation surveillance, disaster assessment, and maritime observation [6], [7], [8], [9].

Usually, the SAR-GMTI systems are generally classified into single-channel and multichannel systems. The fundamental principle of single-channel SAR-GMTI is to utilize the Doppler spectral shift information caused by the target motion to achieve the target detection, which is more suitable for fast-moving and strong target detection outside the main-lobe clutter regions [10], [11]. However, due to the platform's high velocities, the ground clutter spectrum may exhibit severe Doppler broadening because of the different observation angles of ground scatterers relative to the radar line-of-sight [12], causing slow and weak moving target echoes to be submerged in the clutter signals [13], [14]. Effective main-lobe clutter suppression therefore becomes a critical factor influencing the GMTI performance. Compared to single-channel systems, multichannel SAR-GMTI systems provide an additional degree of freedom along the spatial dimension, allowing for optimal matching of the target signal energy while realizing ground clutter rejection by using the multidimensional signal processing technique [15]. Typical multichannel SAR-GMTI methods include displaced phase center antenna (DPCA) [16], [17] and along-track interferometry (ATI) [18], requiring the stringent radar system design. The modified DPCA extends its application condition by registering and interpolating the imaging results of two-channel clutter returns, so as to equalize clutter information [17]. However, these methods are nonadaptive and sensitive to the multichannel radar system errors [19].

To effectively improve the clutter suppression performance in complex scenes, an improved DPCA method is proposed in [20] and [21], which extends the signal processing of a dual-channel system to multichannel adaptive signal processing. The basic principle is to maximize the target output signal-to-clutter-plus-noise ratio (SCNR), estimating adaptive weighting coefficients among all the receiving channels for each SAR pixel to minimize the clutter energy and enhance the target energy. Cerutti-Maori and Sikaneta [22] further proposed the extended DPCA (EDPCA) method, which constructs a 2-D matching function based on different motion parameters for target imaging, and then performs covariance matrix estimation and clutter suppression in the multichannel SAR image domain, dramatically increasing the computational complexity. The joint pixel EDPCA [23] improves the construction precision of the multichannel data vector, using more information to enhance clutter suppression performance. These adaptive clutter suppression methods require a large number of independent and identically distributed (i.i.d.) training samples to accurately estimate the CCM [24], [25]. However, in many practical applications, an observation scene is usually an inhomogeneity scene, such as the junction regions of urban and water areas, causing the lack of enough i.i.d. samples [26], [27]. To address these issues, Rabideau and Steinhardt [28] proposed the power selection training method to improve the inhomogeneity clutter rejection performance, which selects the training samples with the relatively higher power for CCM estimation, so as to form the optimal filter notch depth. Chen et al. [29] proposed the generalized inner product (GIP) criterion to precisely estimate the CCM by removing significantly deviating samples through statistical probability distributions. All these methods realize the clutter rejection via the adaptive processing in the space–time domain, where the distinguishability of clutter components and target signals may still be insufficient because of the restricted available degrees of freedom.

In response to the aforementioned issues, polarization information is introduced as an important means to distinguish the clutter and moving targets, expanding the adaptive processing dimensions and combining the polarization-space–time multidimensional information to enhance the target detection performance [30], [31], [32], [33], [34], [35]. Polarimetric SAR (Pol-SAR) uses horizontal (**H**) and vertical (**V**) polarization modes to collect the radar echoes, and different combinations of polarization transmission and reception modes receive echo signals from the same scattering target, containing the amplitude and phase differences of electromagnetic waves [36], [37]. The combinations of polarization information include **HH**, **HV**, **VH**, and **VV** [38]. Multipolarization and multichannel SAR, offering an additional degree of polarization freedom compared to the traditional multichannel SAR, can effectively improve the clutter rejection and target detection performance [39], [40], [41]. Han et al. [42] proposed an effective CCM estimation method based on online knowledge and polarization knowledge assistance. Yang et al. [43] analyzed the clutter rejection performance by combining spatial and polarization information, effectively improving the clutter suppression performance by extending the signal processing dimension. Considering that polarization domain information is largely independent of target movement speed [44], [45], [46], spatial and polarization domains can be processed independently, so as to improve the computational efficiency.

Motivated by previous works, this article develops a cascaded spatial–polarization processing framework that explicitly exploits polarization-dependent scattering diversity to guide adaptive clutter suppression and target detection. The major contributions of this work can be summarized as follows.

1) We propose a polarization-guided spatial clutter suppression strategy for nonuniform scenes. Rather than estimating the CCM from all available pixels indiscriminately, the scene is first partitioned into scattering-consistent categories using Freeman-Power decomposition and the subsequent Wishart method [47]. Within each category, joint-pixel DPCA (JPDPCA) [4] is combined with GIP-based sample screening to construct a more homogeneous training set, improving the robustness of clutter suppression in heterogeneous backgrounds.
2) We develop an iterative processing strategy for the accurate estimation of target polarization steering vectors and precise coherent synthesis in the polarization domain. Preliminary target detections obtained from single-polarization clutter-suppressed images are first utilized to estimate the target polarization steering vector via eigen-decomposition of the target covariance matrix, based on which the corresponding target

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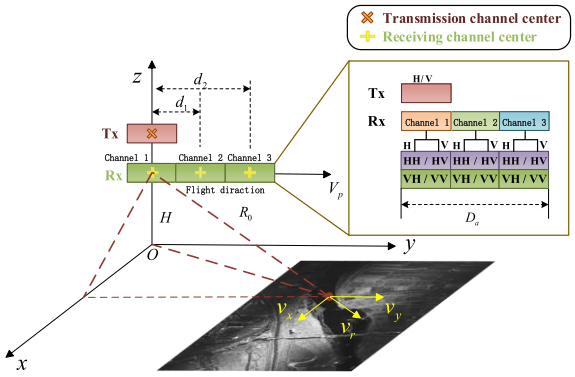


Fig. 1. Spatial arrangement between a multichannel Pol-SAR platform and a GMT.

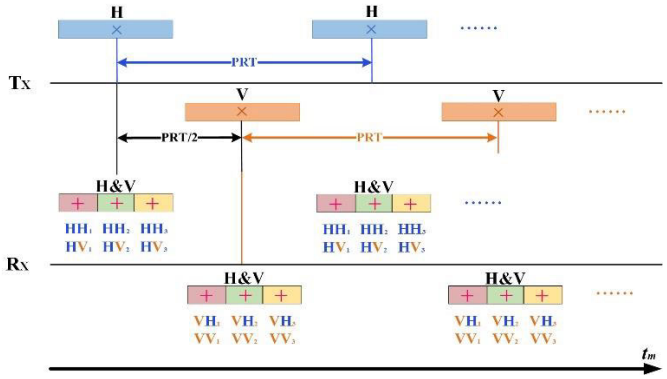


Fig. 2. Airborne polarized SAR interleaved transmitting–receiving model.

polarization projection subspace is constructed. The data from all three polarization channels are then projected onto this subspace. Subsequently, the clutter polarization covariance matrix is re-estimated with the assistance of polarization classification results to derive the optimal adaptive weights for polarization-domain coherent synthesis. This iterative procedure is typically repeated three times to progressively improve the estimation accuracy.

3) We introduce a polarization distance window detector (PD-DT) to suppress false alarms after coherent synthesis. Instead of relying solely on amplitude-driven constant false alarm rate (CFAR) decisions, the proposed detector evaluates the statistical separation between each candidate detection cell and its local background support using the polarization interclass distance. This additional decision layer is particularly effective to improve the GMT detection performance in the complex scene.

The processing results of airborne multipolarization and multichannel real data demonstrate that, compared with traditional methods, the proposed processing chain yields good clutter rejection, high output SCNR, and improved detection probability for moving targets.

The remainder of this article is structured as follows. Section II presents the multichannel Pol-SAR signal model, the real airborne dataset, and the associated preprocessing procedures, including along-track baseline estimation, SAR imaging, multichannel data preprocessing, and polarization-power classification. Section III details the proposed multipolarization and multichannel SAR-GMTI processing framework, including polarization-classification-assisted GIP-weighted JPDPCA clutter suppression, polarization-domain coherent synthesis, and the proposed polarization-based moving target detection strategy. Section IV demonstrates the polarimetric GMTI experimental processing results and discusses the feasibility of the proposed method in detail. Finally, some conclusions are drawn in Section V.

A list of principal symbols is provided in Nomenclature.

II. Multichannel Pol-SAR Signal Model and Signal Preprocessing

A. Multichannel Pol-SAR System Introduction

As shown in Fig. 1, this article discusses a multichannel Pol-SAR GMTI experiment conducted with real data acquired in April 2017 using the airborne N-SAR system [48]. The radar is fully polarimetric and operates in side-looking mode with one single-channel transmitting and multichannel receiving model, operating in the X-band. The aircraft moves along the azimuth direction with the platform speed of V_p, where the platform height is H and the azimuth antenna length is D_a. The full aperture includes three subaperture receiving channels, with the equivalent phase centers of the transmitter and receiver are denoted as d_n, $n = 0, 1, 2$.

The transmitter alternately transmits horizontal and vertical polarization pulses. Each receiving channel can simultaneously receive **H** and **V** polarization echoes. When the transmitting channel chooses the horizontal polarization pulses for transmission, i.e., **H** polarization, each subaperture receiving channel receives the returns with **H** polarization to form the **HH** component, and with **V** polarization to form the **HV** component. Similarly, when the transmitting channel uses the vertical polarization pulses, i.e., **V** polarization, each subaperture receiving channel forms **VH** and **VV** components. A total of four polarization echo signals (**HH**, **HV**, **VH**, and **VV**) can be formed. Due to the reciprocity theorem of polarization, this airborne experimental system only recorded **HH**, **HV**, and **VV** polarization channel data, representing full polarization information.

Fig. 2 shows the scheme of the three-channel fully polarized radar system operation. The transmitter alternately transmits **H** and **V** pulses, and the delays are assumed the same for **HH/HV** and **VH/VV** at the radar receiver. Assuming that the transmitter first emits an **H** polarization pulse, it will then transmit a **V** polarization pulse after a delay of *PRT/2*, where *PRT* represents the pulse repetition interval between two consecutive transmissions of the same polarization.

The experimental data used in this article were collected during a polarized SAR-GMTI experiment conducted in April 2017 in a region to the northeast of the urban area of Weinan City, Shaanxi Province, China. The aircraft flew from south to north along the flight path at a speed of about 128 m/s, with the flight path oriented 30° northeast. The range and azimuth resolutions of this experimental SAR system are approximately 0.5 m. As shown in Fig. 3, the data studied in the experiment includes a scene over the Wei River Bridge, containing multiple ground moving vehicle targets,

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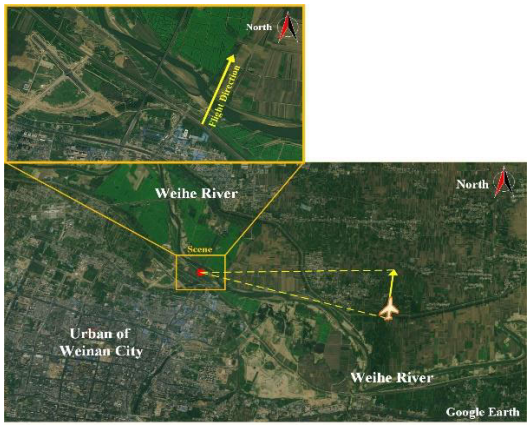


Fig. 3. Relationship between the radar platform and the scene distribution (optical images obtained from Google Earth).

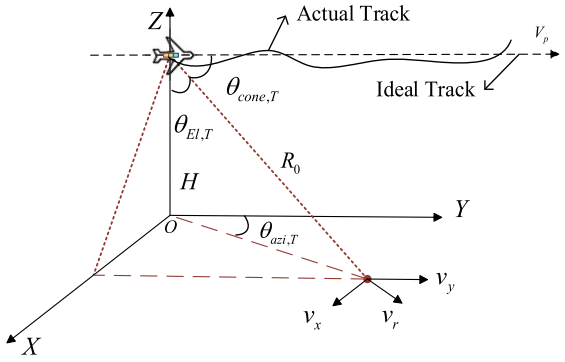


Fig. 4. Multichannel moving target signal model in an aircraft platform with unstable motions.

complex urban buildings, rivers, farmlands, and parts of highways.

B. Airborne Multichannel Pol-SAR GMT and Clutter Echo Model

Fig. 4 illustrates the observation configuration of GMTs as received by the radar on the nonstationary aircraft, where the black solid line represents the actual flight path of the aircraft, and the gray dashed line represents the ideal flight path. The position of the transmitting channel is indicated by $(\Delta X, \Delta Y + V_P t_m, H - \Delta Z)$, where $\Delta X, \Delta Y$, and ΔZ represent the deviations of the aircraft from the ideal flight path along the range, azimuth, and altitude directions, respectively. The position of the GMT can be represented as $(x_0 + v_x t_m, y_0 + v_y t_m, 0)$, where v_x and v_y denote, respectively, the range and azimuth motion velocities of the GMT. The symbol t_m represents the azimuth slow time.

According to the aforementioned model, the slant range history for the nth receiving channel relative to the GMT is expressed as

$$\begin{aligned} R_{MT,n}(t_m) &= [R_T(t_m) + R_n(t_m)]/2 \\ &\approx \sqrt{\begin{array}{l}(x_0 + v_x t_m - \Delta X)^2 + (H - \Delta Z)^2 \\ + \left[\left(V_P - v_y\right) t_m + \Delta Y + d_n/2 - y_0\right]^2\end{array}} \end{aligned} \tag{1}$$

where $R_T(t_m)$ represents the slant range from the transmit channel to the GMT, and $R_n(t_m)$ represents the slant range from the nth receiving channel to the observed target.

After performing the second-order Taylor series expansion on (1), we have

$$\begin{aligned} &R_{MT,n}(t_m) \\ &\approx R_0 + \left[\left(V_P - v_y\right) t_m + \Delta Y + d_n\Big/2\right]^2 \Big/ (2R_0) \\ &\quad + (v_x t_m - \Delta X)^2 \Big/ (2R_0) \\ &\quad - \left[\left(V_P - v_y\right) t_m + \Delta Y + d_n\Big/2\right] y_0 \Big/ R_0 \\ &\quad + (v_x t_m - \Delta X)\, x_0 \Big/ R_0 - \Delta Z \cdot H \Big/ R_0 + \Delta Z^2 \Big/ (2R_0) \\ &\approx R_0 - \left(d_n\Big/2 + \Delta Y\right) \cos\theta_{cone,T} \\ &\quad - \Delta X \sin\theta_{Azi,T} \sin\theta_{El,T} - \Delta Z \cos\theta_{El,T} \\ &\quad + \left[v_r - V_P \cos\theta_{cone,T} + \left(V_P - v_y\right) d_n \Big/ (2R_0)\right] t_m \\ &\quad + \left[\left(V_P - v_y\right) \Delta Y - v_x \Delta X\right] \Big/ R_0 \cdot t_m \\ &\quad + \left[v_x^2 + \left(V_P - v_y\right)^2\right] t_m^2 \Big/ (2R_0) \end{aligned} \tag{2}$$

where $v_r = (x_0 v_x + v_y y_0)/R_0$ represents the radial velocity of the GMT. $\theta_{Azi,T}$, $\theta_{El,T}$, and $\theta_{cone,T}$ represent, respectively, the azimuth angle, depression angle, and spatial cone angle of this GMT relative to the airborne radar. $R_0 = (x_0^2 + y_0^2 + H^2)^{1/2}$ represents the target initial slant range.

Based on (2), the echo signal from the moving target at the nth receiving channel is expressed as

$$\begin{aligned} &s_{MT,n}(t_r, t_m) \\ &= \sigma_{MT}\, rect\left\{\left[t_r - 2R_{MT,n}(t_m)\Big/c\right]\Big/T_p\right\} \\ &\quad \times rect\left[(t_m - t_0)\Big/T_a\right] \exp\left\{j\pi K_r \left[t_r - 2R_{MT,n}(t_m)\Big/c\right]^2\right\} \\ &\quad \times \exp\left[-j4\pi R_{MT,n}(t_m)\Big/\lambda\right] \end{aligned} \tag{3}$$

where t_r, T_p, T_a, K_r, σ_{MT}, $\omega_{azi}(\cdot)$, and $t_0 = (y_0 - d_n/2)/V_P$ represent the fast time, the pulse duration, the synthetic aperture time, the fast-time chirp rate, the scattering amplitude of the moving target, the azimuth window function, and the center time of the aperture, respectively. And $\text{rect}(t/T_p)$ is a rectangle window function defined as 1 within $|t/T_p| \le 0.5$ and 0 elsewhere.

As for the stationary ground clutter, the velocities along-track and perpendicular to the ground track are 0, i.e., $v_x = v_y = 0$; hence the slant range history of a ground static clutter scatterer is

$$\begin{aligned} &R_{C,n}(t_m) \\ &= \sqrt{\begin{array}{l}(x_0 - \Delta X)^2 + (H - \Delta Z)^2 \\ \left(V_P t_m + \Delta Y + d_n/2 - y_0\right)^2\end{array}} \\ &\approx R_0 - \left(\Delta Y + d_n\Big/2\right) \cos\theta_{cone,c} - \Delta X \sin\theta_{Azi,c} \sin\theta_{El,c} \\ &\quad + \left[V_P\left(d_n + 2\Delta Y\right)\Big/(2R_0) - V_P \cos\theta_{cone,c}\right] t_m \\ &\quad - \Delta Z \cos\theta_{El,c} + V_P^2 t_m^2 \Big/ (2R_0) \end{aligned} \tag{4}$$

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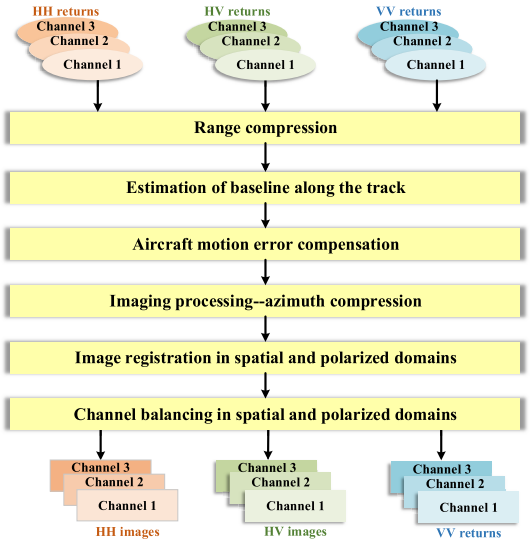


Fig. 5. Preprocessing of multichannel Pol-SAR data workflow.

where $\theta_{Azi,c}$, $\theta_{El,c}$, and $\theta_{cone,c}$ represent, respectively, the azimuth, depression, and spatial cone angle of the clutter scattering point relative to the radar receiver. According to (4), the clutter signal echo at the nth receiving channel is given by

$$\begin{aligned} &s_{C,n}(t_r,t_m)\\ &=\sigma_c rect\left\{\left[t_r-2R_{C,n}(t_m)\Big/c\right]\Big/T_p\right\}\omega_{azi}(t_m-t_0)\\ &\quad\times\exp\left\{j\pi K_r\left[t_r-2R_{C,n}(t_m)\Big/c\right]^2\right\}\\ &\quad\exp\left[-j4\pi R_{C,n}(t_m)\Big/\lambda\right] \end{aligned} \tag{5}$$

where σ_c and λ denote, respectively, the clutter scattering amplitude and the transmitted signal wavelength.

Due to the differences in the slant range histories between moving targets and clutter components, the received echo signals exhibit distinct phase distributions, which causes discrepancies in the corresponding space–time steering vectors, forming the basis for target detection and relocation.

Before proceeding with GMTI processing, it is necessary to perform the data preprocessing, as illustrated in Fig. 5. To better realize the high-precision target radial velocity estimation, the long-track baseline of this SAR system should be precisely estimated. Meanwhile, the aircraft motion errors should be compensated before SAR imaging. The multichannel imaging results are then registered and error-compensated to enhance the clutter data coherence among different receiving channels, thereby laying the foundation for the subsequent reliable target detection and localization.

C. Along-Track Baseline Estimation

Due to the factors such as airflow fluctuations and aircraft yaw, the difference in the equivalent phase centers among the actual radar channels may not be $d = D_a/N$, where N represents the number of receiving channels. Therefore, it is necessary to estimate the along-track baseline [49] based on the radar echo data.

The clutter echo signal after range pulse compression is given by

$$\begin{aligned} &s_{C,n,RC}(t_r,t_m)\\ &=\sigma_c\mathrm{sinc}\left\{B_a\left[t_r-2R_{C,n}(t_m)\Big/c\right]\right\}\\ &\quad\times\mathrm{rect}\left[(t_m-t_0)\Big/T_a\right]\exp\left[-j2\pi V_P^2t_m^2\Big/(\lambda R_0)\right]\\ &\quad\times\exp\left[-j4\pi\left(R_0-d_n\cos\theta_{cone,c}\Big/2\right)\Big/\lambda\right]\\ &\quad\times\exp\left[j4\pi\begin{pmatrix}\Delta X\sin\theta_{Azi,c}\sin\theta_{El,c}+\Delta Y\cos\theta_{cone,c}\\+\Delta Z\cos\theta_{El,c}\end{pmatrix}\Big/\lambda\right]\\ &\quad\times\exp\left\{-j4\pi\left[\begin{matrix}V_Pd_n\Big/(2R_0)+V_P\Delta Y\Big/R_0\\-V_P\cos\theta_{cone,c}\end{matrix}\right]t_m\Big/\lambda\right\} \end{aligned} \tag{6}$$

where B_a is the signal azimuth bandwidth and $\mathrm{sinc}\{x\} = \sin\pi x/\pi x$ is the sinc function.

After performing the azimuth fast Fourier transform (FFT) on (6), one has

$$\begin{aligned} &S_{C,n,RC}(t_r,f_a)\\ &\approx\sigma_c'\mathrm{sinc}\left\{B_a\left[t_r-2R_{C,n}(f_a)\Big/c\right]\right\}\\ &\quad\times\exp\left[-j4\pi\left(R_0-d_n\cos\theta_{cone,c}\Big/2\right)\Big/\lambda\right]\\ &\quad\times\mathrm{sinc}\left[T_a\begin{pmatrix}f_a+V_Pd_n\Big/(\lambda R_0)+2V_P\Delta Y\Big/(\lambda R_0)\\-2V_P\cos\theta_{cone,c}\Big/\lambda\end{pmatrix}\right]\\ &\quad\times\exp\left[j4\pi\begin{pmatrix}\Delta X\sin\theta_{Azi,c}\sin\theta_{El,c}+\Delta Y\cos\theta_{cone,c}\\+\Delta Z\cos\theta_{El,c}\end{pmatrix}\Big/\lambda\right]\\ &\quad\times\exp\left\{-j4\pi\left[\frac{V_Pd_n}{2R_0}+\frac{V_P\Delta Y}{R_0}-V_P\cos\theta_{cone,c}\right]t_0\Big/\lambda\right\}\\ &\quad\times\exp\left\{-j2\pi f_at_0\Big/\lambda\right\}\otimes\exp\left[j\pi\lambda R_0f_a^2\Big/\left(2V_P^2\right)\right]. \end{aligned} \tag{7}$$

Substituting $t_0 = (y_0 - d_n/2)/V_P$ into (7) and omitting the negligible term $\exp[j4\pi d_n^2/(\lambda R_0)]$, one has

$$\begin{aligned} &S_{C,n,RC}(t_r,f_a)\\ &\approx\sigma_c'\mathrm{sinc}\left\{B_a\left[t_r-2R_{C,n}(f_a)\Big/c\right]\right\}\\ &\quad\times\exp\left(-j4\pi R_0\Big/\lambda\right)\exp\left[j2\pi d_n\Delta Y\Big/(\lambda R_0)\right]\\ &\quad\times\mathrm{sinc}\left[T_a\begin{pmatrix}f_a+V_Pd_n\Big/(\lambda R_0)+2V_P\Delta Y\Big/(\lambda R_0)\\-2V_P\cos\theta_{cone,c}\Big/\lambda\end{pmatrix}\right]\\ &\quad\times\exp\left[j4\pi\begin{pmatrix}\Delta X\sin\theta_{Azi,c}\sin\theta_{El,c}+\Delta Y\cos\theta_{cone,c}\\+\Delta Z\cos\theta_{El,c}\end{pmatrix}\Big/\lambda\right]\\ &\quad\times\exp\left\{-j2\pi y_0\left[f_a+V_Pd_n\Big/(\lambda R_0)\right]\Big/V_P\right\}\\ &\quad\times\exp\left\{-j2\pi y_0\left[\begin{matrix}2V_P\Delta Y\Big/(\lambda R_0)\\-2V_P\cos\theta_{cone,c}\Big/\lambda\end{matrix}\right]\Big/V_P\right\}\\ &\quad\times\exp\left(j\pi d_nf_a\Big/V_P\right)\otimes\exp\left[j\pi\lambda R_0f_a^2\Big/\left(2V_P^2\right)\right] \end{aligned} \tag{8}$$

where f_a represents the Doppler frequency and σ_c' denotes the signal amplitude in t_r-f_a domain. And $R_{C,n}(f_a)$ represents the mapping of $R_{C,n}(t_m)$ from the time domain to the frequency domain, which can be obtained by applying the principle of stationary phase. The symbol $\otimes$ denotes the convolution

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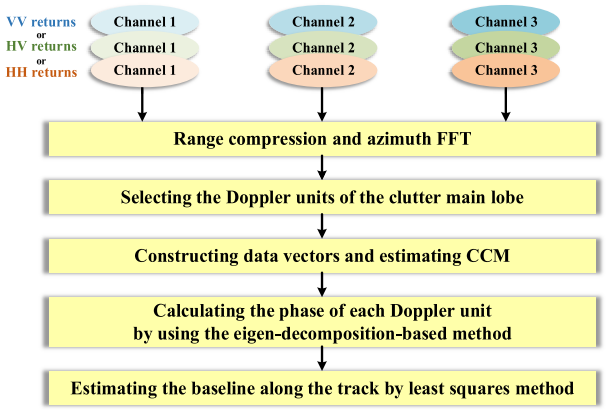


Fig. 6. Flowchart for the along-track baseline estimation by using the eigen-decomposition-based method.

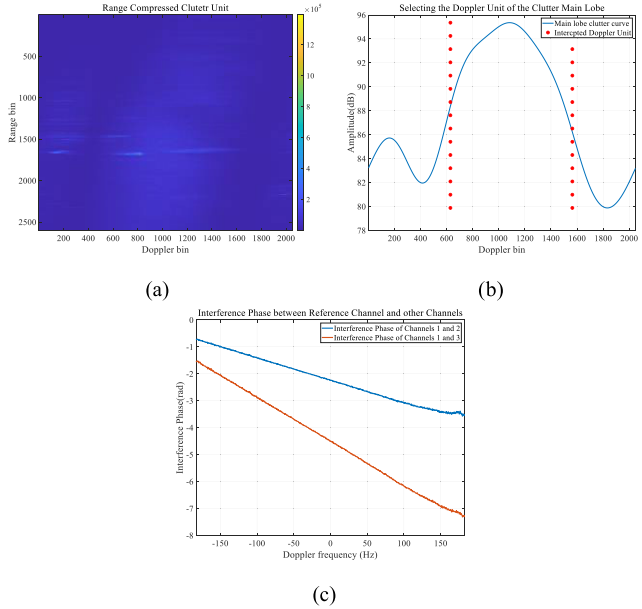


Fig. 7. Experimental results of the along-track baseline estimation. (a) Clutter spectrum distribution after range compression and azimuth Fourier transform. (b) Blue curve represents the clutter distribution across each Doppler cell, with the red portion indicating the range of main-lobe clutter extraction. (c) Interferometric phase distributions between the reference channel and other channels.

operation in the Doppler domain. In (8), the phase term $\exp(j\pi d_n f_a / V_P)$ is crucial to realize the along-track baseline estimation, with the interferometric phase between two receiving channels being

$$\varphi_{n,n-1} = angle\left\{S_{C,n,RC}(t_r, f_a) \cdot S^*_{C,n-1,RC}(t_r, f_a)\right\}. \quad (9)$$

After performing the platform motion error compensation based on the position and orientation system (POS) data, the influence of unstable platform errors on the channel interferometric phase can be effectively alleviated. Then the along-track baseline can be well estimated using the eigen-decomposition-based method in [50], with the processing flowchart provided in Fig. 6.

Following the above process, the real-data processing results are illustrated in Fig. 7. As observed from Fig. 7(a), the

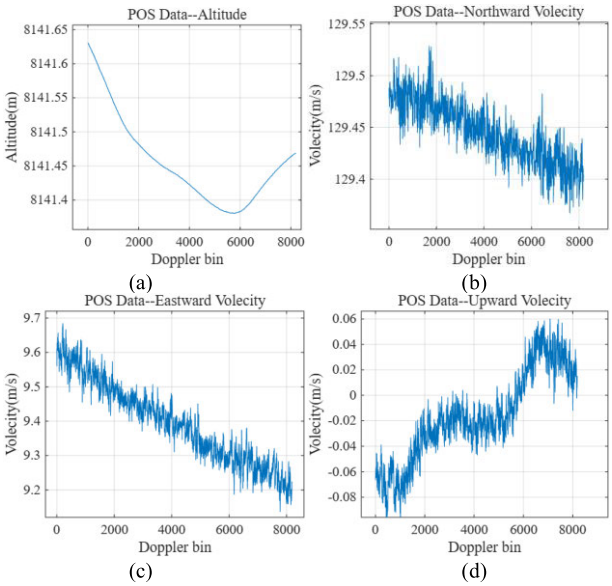


Fig. 8. POS data of this practical SAR system. (a) Aircraft flight altitude data recorded by POS data. (b) Aircraft flight speed data in the northward direction recorded by POS data. (c) Aircraft flight speed data in the eastward direction recorded by POS data. (d) Aircraft flight speed data in the upward direction recorded by POS data.

ground clutter components are severely broadened due to the platform's high-speed motion, forming the main-lobe clutter regions and degrading the relatively slow-moving target detection performance. After averaging the amplitudes along the range dimension, Fig. 7(b) shows the clutter energy distribution for each Doppler gate, representing the weighting information related to the radar antenna pattern. Two vertical circled lines in Fig. 7(b) represent the automatically computed 3-dB main-lobe clutter regions. The region between these two lines contains the dominant main-lobe clutter energy, which is optimal for baseline estimation and terrain phase compensation. According to Fig. 7(b), the range samples located at the main-lobe clutter components are selected for the CCM construction. Then, the interferometric phases between the first and second channels, as well as between the first and third channels, are calculated, and the results are separately shown in Fig. 7(c), from which it can be seen that two straight lines are formed. After performing the least squares method, the along-track baseline is estimated as: $[0, -0.3183, -0.6636]$, coinciding with the nominal values.

D. SAR Imaging Processing

In the following, the SAR imaging processing is performed on the multipolarization and multichannel data, utilizing the SPECtral ANalysis (SPECAN) algorithm [51], [52] to achieve efficient azimuth focusing. The high-precision POS data, recorded by the onboard inertial navigation system during the flight, including altitude and velocity information, serve as the motion compensation reference, as illustrated in Fig. 8.

Based on the aircraft's POS data, the motion error compensation and the SAR imaging processing are performed on the echo data, with the workflow depicted in Fig. 9.

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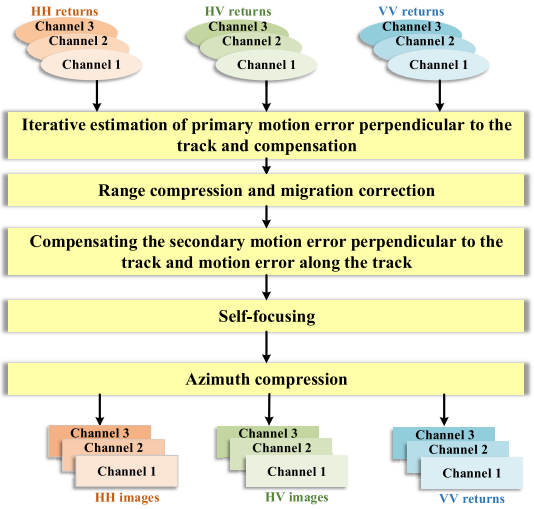


Fig. 9. SAR motion error compensation and imaging processing workflow.

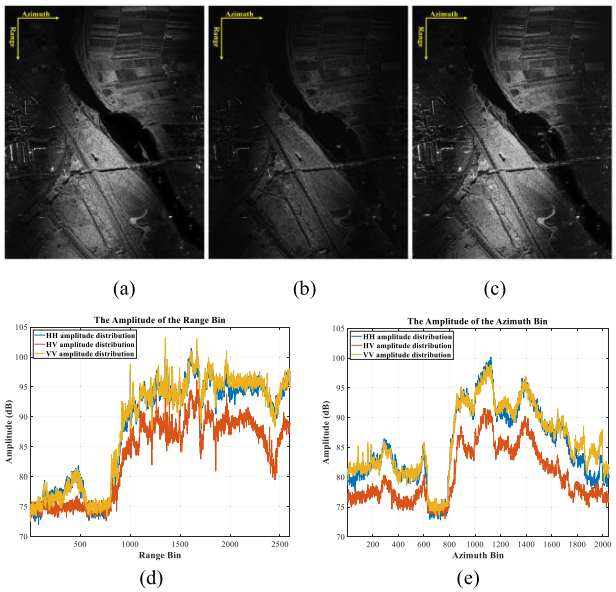


Fig. 10. SAR imaging and clutter power analysis results of **HH**-, **HV**-, and **VV**-channel data. (a) Imaging result of **HH** polarization data (the first azimuth channel). (b) Imaging result of **HV** polarization data (the first azimuth channel). (c) Imaging result of **VV** polarization data (the first azimuth channel). (d) Amplitude distribution from imaging results corresponding to **HH**, **HV**, and **VV** polarizations. (e) Amplitude distribution from imaging results corresponding to **HH**, **HV**, and **VV** polarizations.

It should be noted that, after implementing the motion error compensation, the GMT echo signal can be represented as [53], [54], [55], [56], [57]

$$\begin{aligned} & s'_{MT,n,RC}(t_r, t_m) \\ & \approx \sigma_{MT} \mathrm{sinc}\left\{B_a\left[t_r - 2R_{MT,n}(t_m)/c\right]\right\} \\ & \times rect\left[(t_m - t_0)/T_a\right] \\ & \quad \exp\left[-j4\pi\left(R_0 - d_n\cos\theta_{cone,T}/2\right)/\lambda\right] \\ & \times \exp\left\{-j4\pi\left[v_r - V_P\cos\theta_{cone,T}\right.\right. \\ & \qquad \left.\left. + (V_P - v_y)\, d_n/(2R_0)\right] t_m/\lambda\right\} \\ & \times \exp\left\{-j2\pi\left[v_x^2 + (V_P - v_y)^2\right] t_m^2/(\lambda R_0)\right\}. \end{aligned} \tag{10}$$

Since the ground target motion speed is much smaller compared with the airborne SAR platform, the prior parameters of the radar system can be used to approximately achieve target imaging in (10). When the motion influence of the ground target cannot be ignored, the chirp-parameter-estimation-based methods can be used to realize the target Doppler parameter estimation and focusing. Interested readers may refer to [4] and [53].

Based on the estimated along-track baseline, after azimuth registration, the focused imaging result of the GMT signal can be noted as

$$\begin{aligned} & S'_{MT,n2}(t_r, f_a) \\ & \approx \sigma_{MT2} \mathrm{sinc}\left\{B_a\left[t_r - 2R_{MT,n}(f_a)/c\right]\right\} \\ & \times \mathrm{sinc}\left[T_a\left(f_a + 2v_r/\lambda - 2V_P\cos\theta_{cone,T}/\lambda\right)\right] \\ & \times \exp\left[-j2\pi t_0\left(f_a + 2v_r/\lambda - 2V_P\cos\theta_{cone,T}/\lambda\right)\right] \\ & \times \exp\left(j2\pi d_n\cos\theta_{cone,T}/\lambda\right) \end{aligned} \tag{11}$$

where σ_{MT2} represents the amplitude of the GMT signal after azimuth pulse compression gain.

Similarly, the focused imaging result of the clutter signal in the Doppler domain is given by

$$\begin{aligned} & S_{C,n2}(t_r, f_a) \approx \sigma_{C2} \mathrm{sinc}\left\{B_a\left[t_r - 2R_{C,n}(f_a)/c\right]\right\} \\ & \times \mathrm{sinc}\left[T_a\left(f_a - 2V_P\cos\theta_{cone,c}/\lambda\right)\right] \\ & \times \exp\left[-j2\pi t_0\left(f_a - 2V_P\cos\theta_{cone,c}/\lambda\right)\right] \\ & \times \exp\left(j2\pi d_n\cos\theta_{cone,c}/\lambda\right) \end{aligned} \tag{12}$$

where σ_{C2} represents the amplitude of the clutter signal after azimuth compression.

According to (11), the Doppler center of GMT signal is

$$f_{MT,a0} = 2V_P\cos\theta_{cone,T}/\lambda - 2v_r/\lambda. \tag{13}$$

According to (12), the Doppler center of clutter signal is

$$f_{C,a0} = 2V_P\cos\theta_{cone,c}/\lambda. \tag{14}$$

Based on (11)–(14), after performing the interferometric phase compensation of different Doppler cells by using the proposed spectrum filtering method in [49], the resulting spatial steering vectors for the GMT and clutter are, respectively, given by

$$\mathbf{a}_{s,MT}(v_r) = \left[1, \exp\left(j2\pi\frac{v_r d_1}{\lambda V_P}\right), \exp\left(j2\pi\frac{v_r d_2}{\lambda V_P}\right)\right] \tag{15}$$

$$\mathbf{a}_{s,C} = [1, 1, 1]. \tag{16}$$

According to Fig. 9, the three-channel **HH**, **HV**, and **VV** polarization data are processed, with the imaging results shown in Fig. 10(a)–(c). For the imaging results of **HH**, **HV**, and **VV** polarized data, subblocks from azimuth units 650 to 750 are selected to analyze the energy strength in the range direction,

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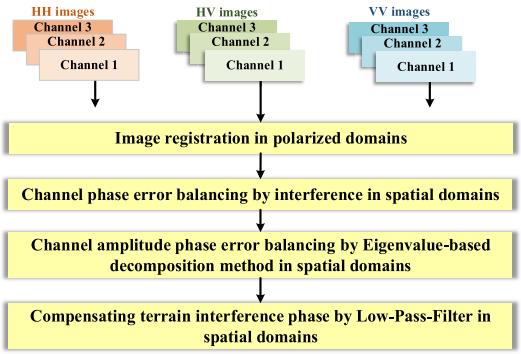


Fig. 11. Channel registration, balancing, and terrain interferometric phase compensation processing workflow.

as shown in Fig. 10(d). In addition, subblocks from range units 650 to 750 are selected to analyze the energy strength in the azimuth direction, as shown in Fig. 10(e). It is observed from Fig. 10(d) and (e) that the amplitudes of **HH** and **VV** polarization are almost identical. In addition, compared to **HV** cross-polarization data, the clutter energy is much weaker, with the power of 5–8 dB lower than that of co-polarized data.

E. Multichannel Data Preprocessing

After performing the channel registration [58], [59] and channel amplitude/phase error compensation [9], [60], the channel coherent coefficients can be effectively improved. However, due to the presence of terrain interferometric phase of clutter components among different receiving channels, the channel coherence is insufficient to accomplish the multichannel clutter rejection and moving target detection, which should be precisely compensated. In [4], according to the distribution property of terrain interferometric phase in the 2-D spectrum domain, the low-pass-filter technique can be applied to efficiently perform the interferometric phase compensation. The overall processing flow is shown in Fig. 11.

As for these multichannel and multipolarization data, after performing the SAR imaging, the interferometric phases between the first and second azimuth receiving channels corresponding to HH, HV, and VV polarization channels are, respectively, depicted in Fig. 12(a)–(c). From the figures, it is observed that the terrain interference phases are significantly deviated from the ideal vector in (16), severely degrading the subsequent adaptive processing performance.

After implementing the channel registration in the polarization domains, applying the amplitude/phase error balancing between spatial channels, and performing the terrain interferometric phase compensation, the interferometric phases with respect to **HH**, **HV**, and **VV** polarization channels are shown in Fig. 12(d)–(f). It can be seen from the figures that the interferometric phases are mainly concentrated on the zero-degree region, which facilitates the subsequent clutter rejection and target radial velocity estimation. It should be noted that the above channel balancing and terrain interferometric phase compensation method cannot be performed along the polarization dimension, since the scene amplitude and phase differences among **HH-**, **HV-**, and **VV-**channels contain the important polarization information.

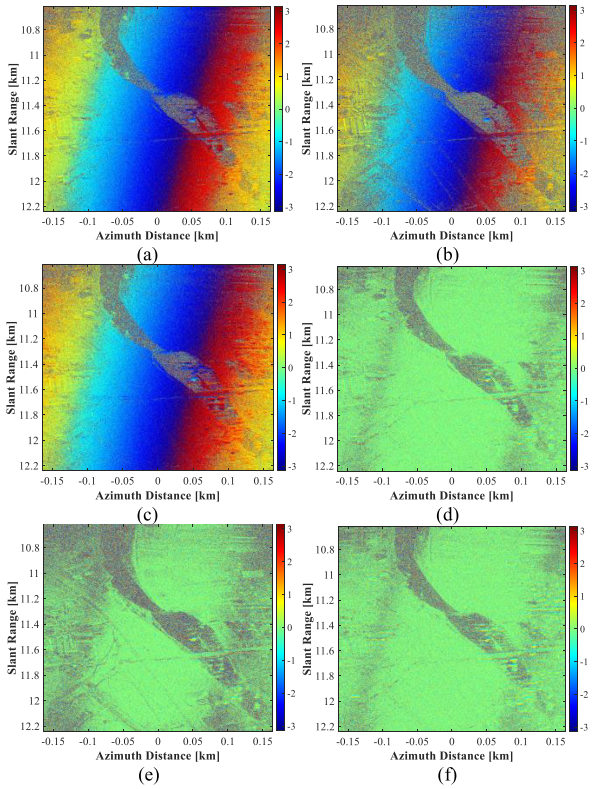


Fig. 12. Multichannel echo data processing. (a) Interferometric phase between the first and second spatial channels with respect to **HH** polarization SAR imaging result. (b) Interferometric phase between the first and second spatial channels with respect to **HV** polarization SAR imaging result. (c) Interferometric phase between the first and second spatial channels with respect to **VV** polarization SAR imaging result. (d) Interferometric phase with respect to **HH** polarization data after implementing SAR imaging, channel registration, channel balancing, and terrain interferometric phase compensation. (e) Interferometric phase with respect to **HV** polarization data after implementing SAR imaging, channel registration, channel balancing, and terrain interferometric phase compensation. (f) Interferometric phase with respect to **VV** polarization data after implementing the SAR imaging, channel registration, channel balancing, and terrain interferometric phase compensation.

According to the processing results in Fig. 12, Fig. 13(a) illustrates the coherence coefficient between the first second spatial channel with respect to **HH** polarization SAR data, and Fig. 13(b) shows the coherence coefficient with respect to the **HH** polarization SAR data after channel registration, channel balancing, and terrain interferometric phase compensation. The histogram statistics results provided in Fig. 13(c) indicate that the channel coherence coefficients can be significantly improved after performing the SAR data preprocessing steps for each polarization channel.

After the aforementioned processing, the clutter distribution properties across the spatial receiving channels with respect to **HH**, **HV**, and **VV** polarization data have essentially become consistent, which is beneficial for the subsequent clutter rejection, moving target detection, and relocation.

F. Polarization-Power Classification

Polarization classification processing exploits the differences in target scattering across multiple polarization channels to classify the observed scene, and the resulting clutter classification can be utilized as prior knowledge for subsequent clutter suppression processing. In this experiment, polarization

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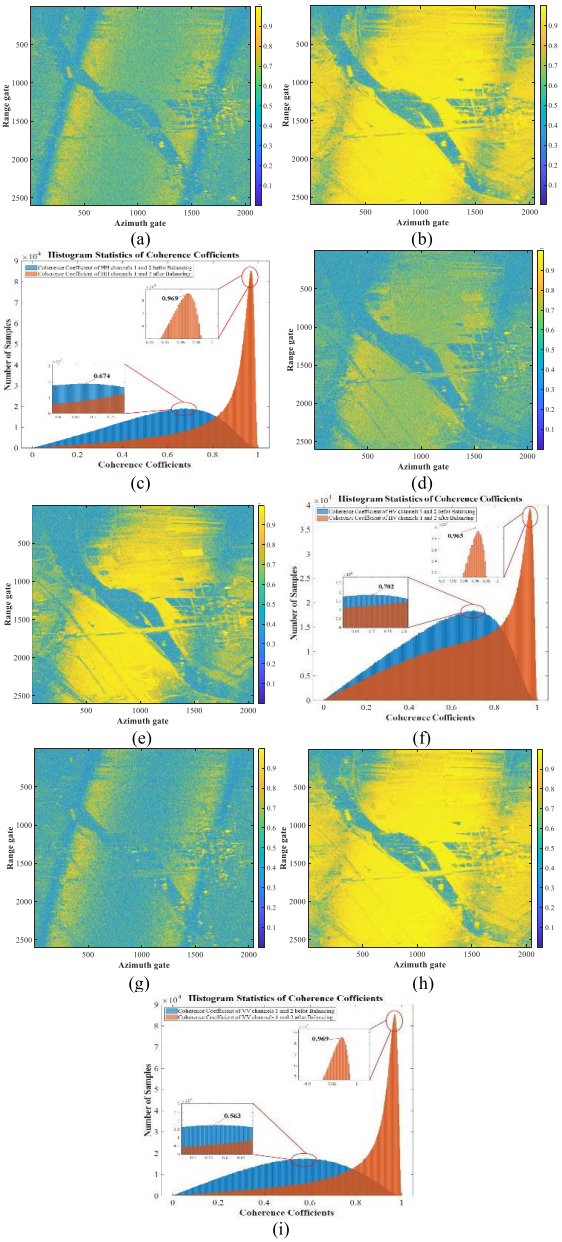


Fig. 13. Coherent coefficient analysis. (a) Coherence coefficient between the first spatial channel and the second spatial channel with respect to the **HH** polarization channel before SAR echo data preprocessing. (b) Coherence coefficient between the first spatial channel and the second spatial channel of the **HH** polarization channel after implementing the channel registration, channel balancing, and terrain interferometric phase compensation. (c) Histogram comparison of (a) and (b). (d) Coherence coefficient between the first spatial channel and the second spatial channel with respect to the **HV** polarization channel before SAR echo data preprocessing. (e) Coherence coefficient between the first spatial channel and the second spatial channel of the **HV** polarization channel after implementing the channel registration, channel balancing, and terrain interferometric phase compensation. (f) Histogram comparison of (d) and (e). (g) Coherence coefficient between the first spatial channel and the second spatial channel with respect to the **VV** polarization channel before SAR echo data preprocessing. (h) Coherence coefficient between the first spatial channel and the second spatial channel of the **VV** polarization channel after implementing the channel registration, channel balancing, and terrain interferometric phase compensation. (i) Histogram comparison of (g) and (h).

classification is performed on the first spatial channel data with respect to the **HH**, **HV**, and **VV** polarization SAR data using

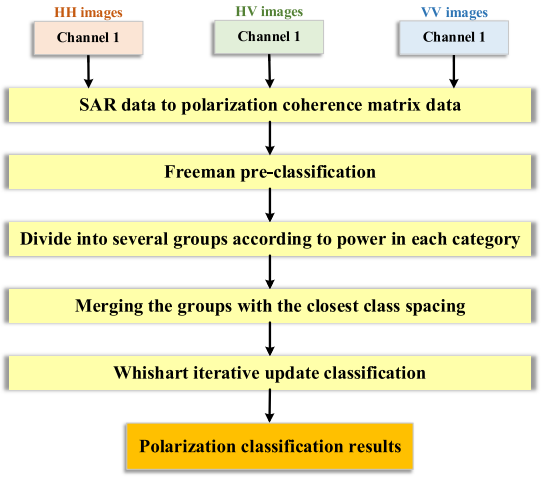


Fig. 14. Polarization-power classification processing workflow.

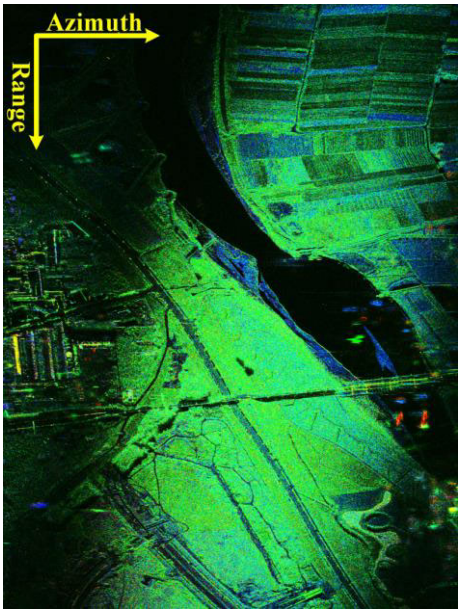


Fig. 15. Freeman decomposition synthesized false-color image, where surface scattering is represented by green, double-bounce scattering is represented by blue, and volume scattering is represented by red.

the Freeman-power decomposition combined with the Wishart iterative method, with the processing flow illustrated in Fig. 14.

According to the Freeman decomposition, the original Pol-SAR image is decomposed into surface scattering, double-bounce scattering, and volume scattering components, and the corresponding synthesized false-color image is shown in Fig. 15.

As can be observed from Fig. 15, river areas are mainly characterized by surface scattering and appear dark in the SAR image due to low backscattering energy. Farmlands and certain ground regions are also dominated by surface scattering and are displayed in green, while some farmland areas are classified as double-bounce scattering because of differences in vegetation structure, appearing in blue. Street lamps, some buildings, and several moving targets are primarily associated with volume scattering and are rendered in red or yellow.

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Based on the dominant scattering mechanism, each pixel is assigned to a corresponding scattering category. Subsequently, the data within each category are further divided according to scattering power magnitude, and the polarization subspaces of each subgroup are calculated and merged. As a result, surface scattering is refined into three subclasses, double-bounce scattering into two subclasses, and volume scattering into three subclasses, yielding a total of eight categories. After implementing the further refinement through Wishart iterative updating, more accurate polarization classification results can be obtained [61].

After completing the aforementioned preprocessing steps, GMT detection and relocation are carried out through joint spatial–polarization domain processing based on the obtained classification results. The detailed polarimetric GMTI processing framework is presented in Section III.

III. Proposed Multipolarization and Multichannel SAR-GMTI Processing Method

Moving target detection is performed based on the results of the preprocessing stage, where moving targets and clutter can be distinguished in both spatial and polarization domains. Under ideal conditions, if the spatial and polarization steering vectors of moving targets are known, optimal adaptive processing can be implemented by matching the target space–time–polarization steering vectors, thereby maximizing the target energy and effectively accomplishing the clutter rejection. However, in practical scenarios, optimal adaptive processing is challenging to be achieved due to target unknown radial velocities and polarization information. In addition, full-dimensional joint adaptive processing in both polarization and spatial domains suffers from high computational complexity and requires a large number of i.i.d. training samples, making it impractical for real-world applications [19].

To address the above limitations, this work proposes a spatial-polarization cascaded GMTI processing framework without requiring prior knowledge of the target polarization steering vectors. Initially, multichannel clutter suppression in the spatial domain is performed independently on the azimuth subchannels of **HH**, **HV**, and **VV** polarizations, using the JPDPCA method, where the CCM estimation is assisted by polarization classification results combined with GIP weighting. Subsequently, the clutter-suppressed results from three polarization channels are coherently synthesized in the polarization domain by exploiting the inherent polarization differences between the clutter components and the moving target signals. Based on the preliminary target detection results obtained from single-polarization clutter suppression, the target polarization steering vectors and the corresponding polarization projection subspace are estimated to perform polarization-domain coherent synthesis. An iterative processing strategy is then employed to update the detected moving target signals and progressively refine the estimation accuracy of the target polarization steering vectors.

After performing the polarization coherent synthesis, CFAR detection is performed on the synthesized results, and a polarization-distance window detector is further employed for secondary screening to reduce the false alarms. Finally, adaptive processing is conducted on the polarization data of each channel using the estimated target polarization steering vectors, thereby enhancing the target output SCNR. The refined detection results are then used to estimate the target radial velocity and accomplish the target relocation. The specific processing workflow is shown in Fig. 16.

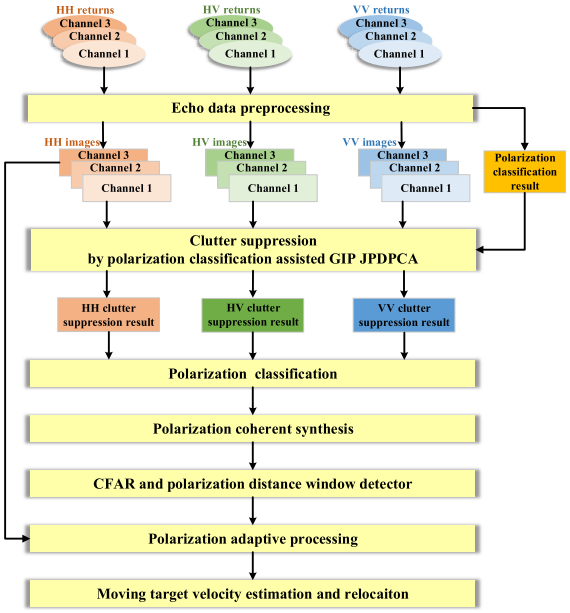


Fig. 16. Multipolarization and multichannel SAR-GMTI processing workflow.

A. Clutter Suppression Processing Based on Polarization Classification-Assisted GIP-Weighted JPDPCA

After preprocessing the echo data, the complex data corresponding to the pixels of the SAR image can be expressed as

$$\begin{aligned} H_0 &: \boldsymbol{x}_s(i) = \boldsymbol{c}_s(i) + \boldsymbol{n}_s(i) \\ H_1 &: \boldsymbol{x}_s(i) = \boldsymbol{s}(i) + \boldsymbol{c}_s(i) + \boldsymbol{n}_s(i) \end{aligned} \tag{17}$$

where H_0 denotes the absence of a moving target, and H_1 denotes the presence of a moving target. The data vector is constructed using a joint multiple-pixel cancellation approach [62], [63], where $\boldsymbol{x}_s(i) \in \mathbb{C}^{L_s \times 1}$ represents the data vector, $\boldsymbol{s}(i) \in \mathbb{C}^{L_s \times 1}$ represents the data vector of the moving target, $\boldsymbol{c}_s(i) \in \mathbb{C}^{L_s \times 1}$ and $\boldsymbol{n}_s(i) \in \mathbb{C}^{L_s \times 1}$, respectively, represent the data vectors of clutter and noise between azimuth subchannels. L_s is the dimension of the data vector.

The joint-pixel data vector for the pixel i is then arranged as

$$\mathbf{X}_{\text{Joint},s}(i) = \left[x_{s,1}(i)\ \mathbf{X}_{J2,s}(i)\ \mathbf{X}_{J3,s}(i) \right]^T \tag{18}$$

where $\mathbf{X}_{J2,s}(i)$ and $\mathbf{X}_{J3,s}(i)$ represent vectors formed by the 1-D arrangement of all pixels within a rectangular window of size $(2l_a + 1) \times (2l_r + 1)$ in the second and third channels, centered at the current pixel i, as shown in Fig. 17.

Clutter suppression essentially involves calculating the optimal weights between subchannels to significantly reduce clutter energy and achieve coherent synthesis of moving target

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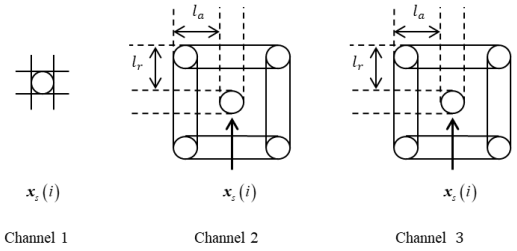


Fig. 17. Schematic of the joint-pixel data vector.

signals, which can be formulated as the following linearly constrained minimum variance optimization problem:

$$\begin{aligned} \min_{\omega_{opt}} \quad & \boldsymbol{\omega}_{opt,s}^{H} \boldsymbol{R}_{cn,s} \boldsymbol{\omega}_{opt,s} \\ \text{s.t.} \quad & \boldsymbol{\omega}_{opt,s}^{H} \boldsymbol{a}_s = 1 \end{aligned} \tag{19}$$

where $\boldsymbol{R}_{cn}$ represents the CCM. The symbols $\boldsymbol{a}_s$ and $\boldsymbol{\omega}_{opt,s}$ represent the GMT steering vector and weighting value, respectively. $(\cdot)^H$ represents the Hermitian transpose.

The traditional CCM estimation method uses all SAR image pixel units to realize the covariance matrix estimation. However, due to the nonuniform characteristics of clutter components, this article estimates the CCM according to different polarization categories, based on the Freeman-Power polarization classification results. It is assumed that the clutter components are divided into M_s polarization categories, with the category weights defined as

$$\begin{cases} \xi_{m_s}(i) = 1, & if \ i \in C_{m_s} \\ \xi_{m_s}(i) = 0, & else \\ where \quad \xi_1(i) + \xi_2(i) + \cdots \xi_{M_s}(i) = 1 \end{cases} \tag{20}$$

where $m_s = 1, 2, \ldots, M_s$ and C_{m_s} represents the training samples contained in the category m_s. First, the initial CCM corresponding to the samples of category m_s is calculated as

$$\hat{\boldsymbol{R}}_{cn,s,0}(m_s) = \frac{1}{N_{C_{m_s}}} \sum_i \xi_{m_s}(i) \cdot \mathbf{X}_{\text{Joint},s}(i) \mathbf{X}_{\text{Joint},s}^{H}(i), i \in C_{m_s} \tag{21}$$

where $N_{C_{m_s}}$ represents the number of training samples contained in the category m_s classification results in the SAR image domain, and $\boldsymbol{x}_s(i), i \in C_{m_s}$ is the SAR data vector of the ith pixel.

Based on the definition of the generalized GIP, the weight of each data sample is calculated as

$$GIP_{m_s}(i) = diag\left(\mathbf{X}_{\text{Joint},s}^{H}(i) \hat{\boldsymbol{R}}_{cn,s,0}^{-1}(m_s) \mathbf{X}_{\text{Joint},s}(i)\right), i \in C_{m_s}. \tag{22}$$

The probability distribution of the GIP samples is then statistically analyzed, and the samples with higher probabilities are selected to compute the averaged GIP sample value

$$GIP_{m_s,av} = \underset{i \in \{P(GIP) \geq p_{m_s}\}}{E} \left[GIP_{m_s}(i)\right] \tag{23}$$

where $p_{m_s} = E[P(GIP_{m_s}(i))]$ with $P(\cdot)$ being the probability distribution function of the GIP.

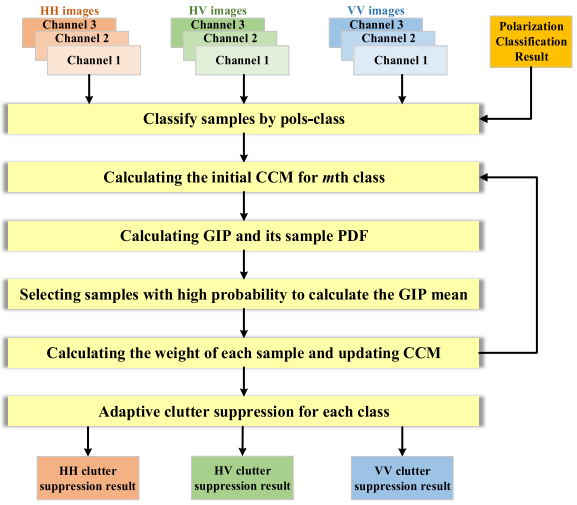


Fig. 18. Processing workflow of the adaptive CCM algorithm using polarization classification and GIP sample screening for multipolarization and multichannel SAR-GMTI system.

The weight for each sample is then defined as [4]

$$\omega_{GIP_{m_s}}(i) = \frac{\exp\left[-\frac{|GIP_{m_s}(i) - GIP_{m_s,av}|}{GIP_{m_s,av}}\right]}{\sum_i \exp\left[-\frac{|GIP_{m_s}(i) - GIP_{m_s,av}|}{GIP_{m_s,av}}\right]}. \tag{24}$$

By weighting the samples, the estimated CCM is obtained as

$$\hat{\boldsymbol{R}}_{cn,s,GIP}(m_s) = \frac{1}{N_{C_{m_s}}} \sum \omega_{GIP_{m_s}}(i) \cdot \xi_{m_s}(i) \cdot \mathbf{X}_{\text{Joint},s}(i) \mathbf{X}_{\text{Joint},s}^{H}(i). \tag{25}$$

Using the updated covariance matrix, the GIP weighting process is recalculated and iteratively refined until the CCM is converged.

Based on the estimated CCM, the clutter rejection weight vector can be obtained as

$$\boldsymbol{\omega}_{opt,s}(m_s) = \frac{\boldsymbol{R}_{cn,s,GIP}^{-1}(m_s) \boldsymbol{a}_s}{\boldsymbol{a}_s^{H} \boldsymbol{R}_{cn,s,GIP}^{-1}(m_s) \boldsymbol{a}_s}. \tag{26}$$

According to (15) and (16), in the absence of the known radial velocity of the moving target, a suboptimal adaptive processing method can be used to ensure the significant suppression of main-lobe clutter components, while achieving coherent synthesis of the moving target with the steering vector of $\boldsymbol{a}_s$ setting as $\boldsymbol{a}_s = [1, 0, 0, \ldots]^T$. According to the above processing based on the polarization classification results, the final result of clutter suppression is given by

$$\mathbf{y}(i) = \sum_{m_s=1}^{M_s} \boldsymbol{\omega}_{opt,s}^{H}(m_s) \left(\xi_{m_s}(i) \mathbf{X}_{\text{Joint},s}(i)\right). \tag{27}$$

Based on the above principles, the clutter suppression processing workflow is shown in Fig. 17.

B. Polarization Domain Coherent Synthesis Algorithm

After applying the clutter suppression processing described in Fig. 18 to each polarization channel, the residual data

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from these three polarization channels are further processed through polarization-domain coherent synthesis. By exploiting the inherent differences between moving targets and clutter components in the polarization domain, this processing stage aims to further suppress the residual clutter components while achieving coherent enhancement of the target signals along the polarization direction.

A polarization-domain data model is established to determine the presence of moving targets in the polarization domain. After clutter suppression, the complex data corresponding to each SAR image pixel in the polarization channels can be expressed as

$$\begin{aligned} H_0 &: \boldsymbol{x}_{res}(i) = \sigma_{c,i}\boldsymbol{s}_{pol,c}(i) + \boldsymbol{n}_{pol}(i) \\ H_1 &: \boldsymbol{x}_{res}(i) = \sigma_{t,i}\boldsymbol{s}_{pol,t}(i) + \sigma_{c,i}\boldsymbol{s}_{pol,c}(i) + \boldsymbol{n}_{pol}(i) \end{aligned} \tag{28}$$

where $\sigma_{t,i}$ and $\sigma_{c,i}$, respectively, represent the amplitudes of the target and clutter signals after spatial domain clutter suppression, with $S_{HH} = |S_{HH}|e^{j\phi_{HH}}$ and $\sigma_{c,i} = |S_{HH}|$. $\boldsymbol{s}_{pol,t} \in \mathbb{C}^{L_p \times 1}$ and $\boldsymbol{s}_{pol,c}(i) \in \mathbb{C}^{L_p \times 1}$ are the target and clutter polarization steering vectors, respectively. $\boldsymbol{n}_{pol}(i) \in \mathbb{C}^{L_p \times 1}$ represents the data vector of noise in the polarization channels after clutter suppression, and $L_p = 3$ represents the number of polarization channels.

As for both target and clutter components, the polarization steering vectors can be expressed as

$$\mathbf{s}_{pol} = \left[\cos\theta_1 \cos\theta_2 \quad \cos\theta_1 \sin\theta_2 e^{j\eta_1} \quad \sin\theta_1 e^{j\eta_1}\right] \tag{29}$$

where $\theta_1 = \arctan(\varepsilon/(1+r^2)^{1/2})$, $\theta_2 = \arctan\sqrt{r}$, with r and ε, respectively, representing the power ratios of **HV** and **VV** to the **HH** channel signals. η_1 and η_2 represent the phase difference between **HV** and **HH** channel signals, and between **VV** and **HH** channel signals, respectively.

Polarization-domain processing can be formulated as an adaptive filtering problem that determines the optimal weighting coefficients among polarization channels to minimize clutter energy while coherently synthesizing the target signal. This process can be described by the minimum variance optimization problem as follows:

$$\begin{aligned} \min_{\omega_{opt}} \quad & \boldsymbol{\omega}_{opt,pol}^H \boldsymbol{R}_{pol,c} \boldsymbol{\omega}_{opt,pol} \\ \text{s.t.} \quad & \boldsymbol{\omega}_{opt,pol}^H \mathbf{s}_{pol,target} = 1 \end{aligned} \tag{30}$$

where $\boldsymbol{R}_{pol,c}$ denotes the clutter polarization covariance matrix, $\boldsymbol{\omega}_{opt,pol}$ represents the polarization-domain adaptive weight vector, and $\boldsymbol{s}_{pol,target}$ is the target polarization steering vector.

In practice scenarios, the polarization steering vectors of moving targets are generally unknown, and multiple targets may coexist within the observed scene. Considering that ground vehicle targets are man-made objects and typically share similar polarization scattering characteristics [64], this article proposes a polarization-domain clutter suppression and target detection algorithm based on coherent synthesis of moving target polarization space.

First, CFAR detection is performed on the clutter-suppressed results from each single-polarization channel. Although the detection results may contain false alarms, the detected pixels generally retain the dominant target signals. Based on the polarization steering vectors extracted from these

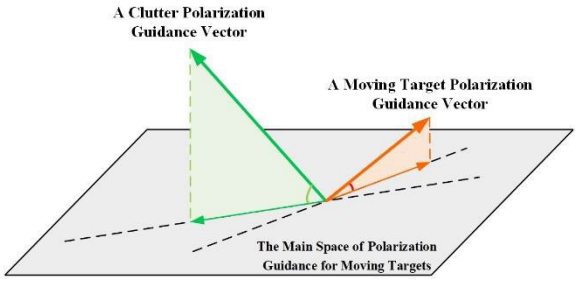


Fig. 19. Clutter and targets in the main polarization space of the target.

detected pixels, the target polarization covariance matrix is constructed as

$$\boldsymbol{R}_{pol,t} = \frac{1}{N_T}\sum_{i=1}^{N_T} \boldsymbol{s}_{pol,t_i} \cdot \boldsymbol{s}_{pol,t_i}^H \tag{31}$$

where $\boldsymbol{s}_{pol,t_i}$ represents the polarization steering vector of the detected moving target points, and N_T represents the number of detected targets.

Eigen-decomposition is performed on (31) as follows:

$$\boldsymbol{R}_{pol,t} = \sum_{i=1}^{L} \beta_i \boldsymbol{v}_i \boldsymbol{v}_i^H \tag{32}$$

where β_i and $\boldsymbol{v}_i$, respectively, represent the eigenvalues and eigenvectors of the covariance matrix. It satisfies $\beta_1 \gg \beta_2 > \beta_3$, and the eigenvector corresponding to the largest eigenvalue is normalized to obtain the estimated target polarization steering vector.

Based on the differences between the polarization steering vectors of targets and clutter components, the amplitude of data projection from each pixel of the entire image onto the subspace of the target polarization steering is calculated. This allows for the rejection of clutter components, while achieving target polarization synthesis, as shown in Fig. 19.

Simultaneously, based on the detected moving targets, the projection subspace of the target polarization steering vector is constructed as

$$\boldsymbol{P}_T = \frac{1}{N_T}\sum_{i=1}^{N_T} \boldsymbol{s}_{pol,t_i}\left(\left(\boldsymbol{s}_{pol,t_i}\right)^H \boldsymbol{s}_{pol,t_i}\right)^{-1}\left(\boldsymbol{s}_{pol,t_i}\right)^H. \tag{33}$$

The data vector after performing the data projection along the polarization direction can be expressed as

$$\begin{aligned} &\boldsymbol{x}_{\text{project}}(i) \\ &= \boldsymbol{P}_T \boldsymbol{x}_{res}(i) \\ &= \left(\frac{1}{N_T}\sum_{i=1}^{N_T} \boldsymbol{s}_{\boldsymbol{s}_{pol,t_i}}\left(\left(\boldsymbol{s}_{\boldsymbol{s}_{pol,t_i}}\right)^H \boldsymbol{s}_{\boldsymbol{s}_{pol,t_i}}\right)^{-1}\left(\boldsymbol{s}_{\boldsymbol{s}_{pol,t_i}}\right)^H\right)\boldsymbol{x}_{res}(i). \end{aligned} \tag{34}$$

Based on the data vector in (32), the clutter's polarization covariance matrix is estimated as

$$\boldsymbol{R}_{pol,c} = \boldsymbol{P}_T\left(\sum_i \boldsymbol{s}_{pol,c}(i) \cdot \boldsymbol{s}_{pol,c}^H(i)\right)\boldsymbol{P}_T^H. \tag{35}$$

To improve the statistical independence and estimation robustness of the polarization CCM, polarization classification

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information is incorporated into the covariance matrix estimation. Specifically, the polarization CCM is estimated based on the clutter suppression outcomes from the **HH**, **HV**, and **VV** polarization channels, which are classified according to the Freeman-power criteria. Analogous to the spatial-domain processing delineated in (20), it is assumed that the clutter can be divided into M_p polarization categories, with the corresponding category weights defined as follows:

$$\begin{cases} \eta_{m_p}(i) = 1, & if \quad i \in C_{m_p} \\ \eta_{m_p}(i) = 0, & else \\ where & \eta_1(i) + \eta_2(i) + \cdots \eta_{M_p}(i) = 1 \end{cases} \tag{36}$$

where $m_p = 1, 2, \ldots, M_p$ and C_{m_p} represents the clutter contained in the polarization category m_p. Then, the polarization CCM for category m_p is given by

$$\hat{\boldsymbol{R}}_{cn,pol}(m_p) = \frac{1}{N_{C_{m_p}}} \sum_{i=1}^{N_{C_{m_p}}} \eta_{m_p}(i) \cdot \left[\boldsymbol{P}_T \boldsymbol{s}_{pol,c}(i)\right] \cdot \left[\boldsymbol{P}_T \boldsymbol{s}_{pol,c}(i)\right]^H \tag{37}$$

where $N_{C_{m_p}}$ represents the number of polarization steering vectors contained in the polarization category m_p.

Based on the estimated polarization steering vector $\boldsymbol{s}_{pol,target}$ and the polarization CCM in (37), the optimal adaptive weight for polarization-domain processing is calculated as

$$\boldsymbol{\omega}_{opt,pol}(m_p) = \frac{\hat{\boldsymbol{R}}_{cn,pol}^{-1}(m_p)\, \boldsymbol{s}_{pol,target}}{\boldsymbol{s}_{pol,target} \hat{\boldsymbol{R}}_{cn,pol}^{-1}(m_p)\, \boldsymbol{s}_{pol,target}}. \tag{38}$$

Applying the adaptive weights to the polarization-domain projected data yields the coherent synthesis output

$$\begin{aligned} \boldsymbol{y}(i) &= \sum_{m_p=1}^{M_p} \boldsymbol{\omega}_{opt,pol}^H(m_p)\, \eta_{m_p}(i)\, \boldsymbol{x}_{\text{project}}(i) \\ &= \sum_{m_p=1}^{M_p} \boldsymbol{\omega}_{opt,pol}^H(m_p)\, \eta_{m_p}(i)\, \boldsymbol{P}_T \boldsymbol{x}_{res}(i). \end{aligned} \tag{39}$$

Considering that the target polarization steering vector estimation result may not be precise, target detection can be performed again based on the results of the polarization coherent synthesis process to ensure accurate selection of moving targets. Using the updated moving target detection results, the target polarization steering vector and polarization projection subspace are updated, and then the polarization coherent synthesis is performed again based on the aforementioned principles. Thus, through iterative processing, a more precise estimation and polarization coherent synthesis of the target signal are achieved.

Based on the aforementioned derivation, the output result of the iterative estimation and polarization coherent synthesis of the target polarization steering vector is

$$\boldsymbol{y}_k(i) = \sum_{m_p=1}^{M_p} \boldsymbol{\omega}_{opt,pol_k}^H(m_p)\, \eta_{m_{pol}}(i)\, P_{T_k} \boldsymbol{x}_{res}(i) \tag{40}$$

where k represents the iteration number and ω_{opt,pol_k} represents the adaptive weights from the kth iteration. P_{T_k} denotes the projection subspace corresponding to the kth iteration target detection results. The polarization steering vector estimated from the kth iteration is denoted as $\boldsymbol{s}_{pol,target_k}$.

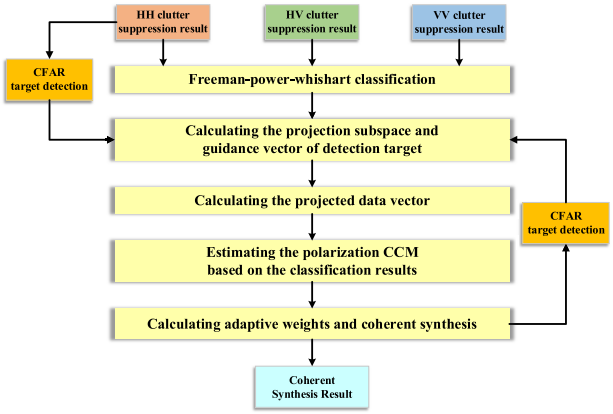


Fig. 20. Polarization coherent synthesis processing workflow.

Based on the above principles, the processing workflow for the proposed polarization coherent synthesis is shown in Fig. 20.

C. Polarization Distance Window Detector

Traditional CFAR detection determines the detection thresholds based on the SCNR of the moving target after clutter suppression. However, in complex scenarios, the powers of clutter scatterers with relatively strong intensions may not be effectively weakened, leading to elevated false alarm rates. Based on the polarization differences between the moving targets and background clutter components, this article extends the conventional CFAR detector framework by implementing a PD-DT.

The polarization coherence matrix is assumed to follow a complex Wishart distribution [47]. Utilizing the maximum likelihood criterion, the Wishart distance measure is defined as

$$d\left([\boldsymbol{T}] \,|\, [\hat{\Sigma}_i]\right) = \ln\left|[\hat{\Sigma}_i]\right| + Tr\left([\hat{\Sigma}_i]^{-1}[\mathbf{T}]\right) \tag{41}$$

where $[\boldsymbol{T}]$, $[\hat{\Sigma}_i]$, $|[\hat{\Sigma}_i]|$, and $Tr(\cdot)$ denote, respectively, the polarimetric coherence matrix of a pixel, the class center of the ith category, the determinant of $[\hat{\Sigma}_i]$, and the trace of a matrix. The class center of the ith category is defined as the sample mean of the coherence matrices within that category

$$[\hat{\Sigma}_i] = E([\boldsymbol{T}]). \tag{42}$$

Accordingly, the interclass distance between the two categories can be expressed as [63]

$$D_{ij} = \frac{1}{2}\left\{\ln\left(\left|[\hat{\Sigma}_i]\right|\right) + \ln\left(\left|[\hat{\Sigma}_j]\right|\right) + Tr\left([\hat{\Sigma}_i]^{-1}[\hat{\Sigma}_j] + [\hat{\Sigma}_j]^{-1}[\hat{\Sigma}_i]\right)\right\}. \tag{43}$$

A larger value of D_{ij} indicates a higher degree of statistical separability between these two classes.

Due to the presence of target radial velocities, their Doppler frequencies deviate from those of stationary clutter components, resulting in distinct polarization characteristics

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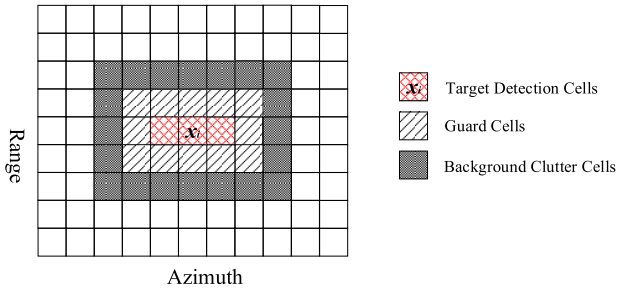


Fig. 21. Pols-distance detection window distribution.

difference with respect to the background clutter scatterers. Consequently, the polarization interclass distance between target and background clutter provides a discriminative feature, enabling effective separation of moving targets from clutter components. A polarization-distance detector is further employed as a postprocessing stage to refine CFAR detection results and suppress false alarms.

The detection procedure is summarized as follows. First, candidate targets identified by CFAR are designated as detection cells. Guard cells are placed around each detection cell to prevent target leakage into the clutter reference region, followed by the selection of background clutter cells surrounding the guard cells. The polarization coherence matrices of the detection cell and the background clutter cells are then estimated. Subsequently, the polarization interclass distance defined in (43) is adopted as the detection threshold value Y. The presence of a target is determined according to the following decision rule:

$$\begin{cases} T \geq Y, & \textit{where no target exists} \\ T < Y, & \textit{target exists} \end{cases} \tag{44}$$

where Y is the polarization interclass difference for the current detection cell and T is the detection threshold. If the detection value Y at the detection cell exceeds the threshold T, it is determined that the moving target signal is present; otherwise, it is determined that no target is present. The setting of the detection threshold can be statistically established based on the background clutter.

Fig. 21 illustrates the schematic distribution of polarization-distance CFAR (Pols-CFAR) detection window. Based on the interclass differences between the target detection cells and background clutter cells, a secondary decision is made for each detected target to reduce the false alarms in the CFAR detection.

IV. Airborne Experiment Processing Results

This section aims to demonstrate that, compared to traditional azimuth multichannel SAR-GMTI systems, multipolarization and multichannel SAR systems combined with the proposed methods described in Section III can significantly enhance the moving target detection performance. The experimental scenario is shown in Fig. 22. Based on the preprocessing results presented in Section II, clutter suppression is first performed independently for each polarization channel, followed by polarization-domain coherent synthesis and polarization-assisted moving target detection. The final localization results illustrate the performance of the proposed approach. A detailed analysis of multichannel real data experiments is discussed below.

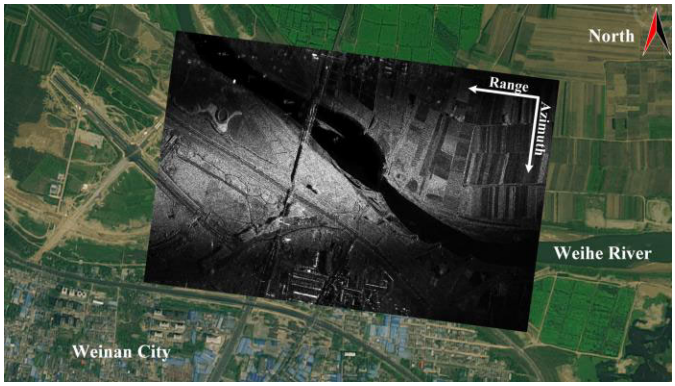


Fig. 22. Experimental scenario of the first **HH** polarization channel SAR imaging data overlaid with Google optical images.

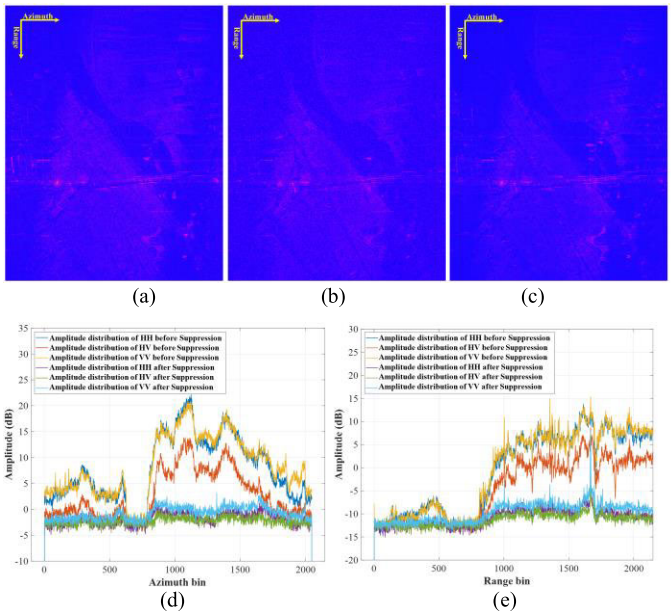
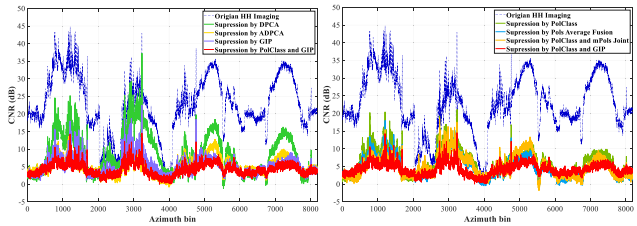


Fig. 23. Multichannel clutter rejection results along spatial domain for **HH**, **HV**, and **VV** polarization channels. (a) Clutter suppression result for the **HH** polarization channel. (b) Clutter suppression results for the **HV** polarization channel. (c) Clutter suppression processing for the **VV** polarization channel. (d) Clutter amplitude averaged along the azimuth dimension before and after multichannel clutter suppression for the **HH**, **HV**, and **VV** polarization channels. (e) Clutter amplitude averaged along the range dimension before and after multichannel clutter suppression for the **HH**, **HV**, and **VV** polarization channels.

A. Spatial-Domain Clutter Suppression Result Analysis

The proposed polarization-assisted classification and GIP-weighted adaptive clutter suppression method is applied in the image domain to suppress clutter separately in three polarization channels. The corresponding results for each polarization channel are shown in Fig. 23(a)–(c). It is found from the figures that after clutter suppression, the **VV** polarization channel retains the highest level of residual clutter energy, while the moving target energy in the cross-polarization **HV** channel is

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(a) (b)

Fig. 24. Clutter rejection performance comparison results. (a) Averaged CNR along the range dimension for the **HH** channel after performing the spatial domain clutter suppression processing. (b) Averaged CNR along the range dimension for the **HH** channel after performing the clutter suppression processing using the polarization information.

lower than that in the co-polarized channels. According to the normalized amplitude comparison results in Fig. 23(d), the cross-polarization channel inherently exhibits the lower signal energy, and the residual clutter energy after clutter suppression remains relatively small. In contrast, the spatial-domain clutter adaptive suppression results for the co-polarization channels are significantly better than those for the cross-polarization **HV** channel.

To demonstrate the effectiveness of the proposed clutter suppression algorithm, which is based on polarization-assisted classification and GIP-weighted JPDPCA, the **HH** polarization channel of the actual SAR data is compared with several representative methods: DPCA, adaptive DPCA (ADPCA), GIP-weighted ADPCA, polarization-assisted ADPCA [46], ADPCA with joint three-polarization channel polarization-assisted classification, and clutter rejection algorithm for averaging fusion across three polarization channels, as shown in Fig. 24(a) and (b). It can be observed from the figures that the proposed algorithm exhibits superior clutter rejection performance. As shown in Fig. 24(a), compared with the GIP-weighted ADPCA algorithm, the proposed approach reduces the clutter-to-noise ratio (CNR) by 1–3 dB in the inhomogeneous clutter scene regions. As shown in Fig. 24(b), compared to polarization-assisted ADPCA, the CNR is reduced by 2–6 dB by using the proposed method.

B. Coherent Synthesis Along Polarization-Channel Domain

Due to the independence of spatial and polarization domains, clutter suppression processing in the spatial domain does not affect the signal synthesis in the polarization domain. In this section, real data are employed to validate the effectiveness of the proposed polarization-domain coherent synthesis method using the clutter spatial-domain suppression results from the three polarization channels.

First, the clutter suppression results are further processed using polarization classification. Fig. 25 illustrates the Freeman decomposition synthesized false-color image derived from the clutter spatial-domain suppression results of the **HH**, **HV**, and **VV** polarization channels. In the figures, one can see that most of the residual ground clutter components belongs to the surface scattering, appearing green in the SAR image. In contrast, the moving targets, which are predominantly man-made vehicles, exhibit stronger backscattered energy,

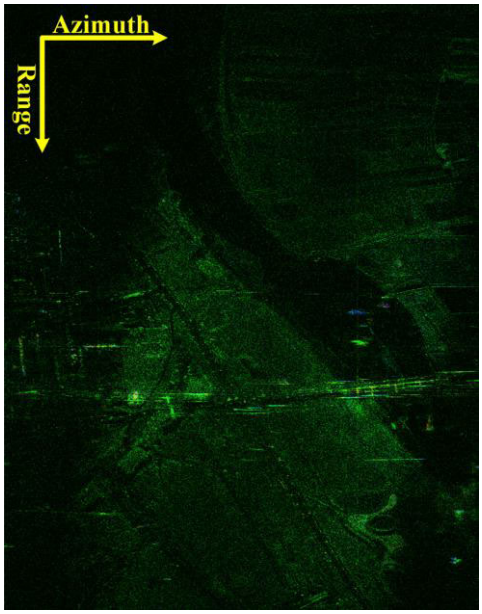


Fig. 25. Freeman decomposition synthesized false-color image of the clutter results from three polarization channels, where surface scattering is represented by green, double-bounce scattering by blue, and volume scattering by red.

and their scattering responses are mainly characterized by volume and double-bounce mechanisms, appearing in red and blue in the SAR image, respectively. Strong residual clutter components, such as buildings and street lights, also manifest as red or blue regions due to similar scattering characteristics.

Then, the cell-averaging CFAR (CA-CFAR) detector [65] combined with the proposed polarization distance based on CFAR method is applied to the clutter-suppressed **HH** polarization image for moving target detection. The corresponding results are presented in Fig. 26(a), where the detection threshold is set as 13 dB. The preliminary target detection results are used to estimate the projection subspace and polarization steering vector of the moving targets. The clutter-suppressed data from all three polarization channels is projected onto the estimated target subspace.

Next, the polarization covariance matrices are estimated according to the polarization classification results, and adaptive processing is conducted in the polarization domain to achieve coherent synthesis across polarization channels. Finally, CA-CFAR detection with the same detection threshold is subsequently reapplied to the polarization coherent synthesis results, followed by re-estimating the moving targets projection subspace and steering vector. The polarization coherent synthesis is iteratively performed again on the three-polarization clutter-suppressed data. Fig. 26(a)–(d) shows the target detection results in each iteration to estimate the target polarization steering vector and projection subspace. The red boxes highlight the moving targets that are not detected in the **HH** polarization channel, and comparison results reveal that the polarization coherent synthesis process can effectively enhance the detection probability of slow moving and weak ground targets.

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TABLE I
SCNRs OF TARGETS T1 TO T30 IN THE THREE POLARIZATION CHANNELS AND THE ITERATIVE RESULTS OF POLARIZATION COHERENT SYNTHESIS

Target Number	**HH**-Pol in dB	**HV**-Pol in dB	**VV**-Pol in dB	Pol-Synthesis once in dB	Pol-Synthesis twice in dB	Pol-Synthesis three-times in dB
No.1	8.6462	25.5191	25.6825	22.0210	28.1660	28.6872
No.2	8.5604	12.8309	17.8156	16.0097	19.8950	20.0151
No.3	9.2244	8.1920	16.9024	15.5976	19.4030	19.3062
No.4	21.3084	23.6834	26.6953	27.7577	30.5683	30.1091
No.5	21.5777	22.1159	27.2043	27.7950	30.7138	30.3003
No.6	5.3761	11.9788	15.3406	14.0742	18.3517	18.3966
No.7	13.7297	14.8726	19.5026	18.6500	22.4626	22.4326
No.8	6.9151	9.6977	19.0795	16.1512	20.7230	20.8701
No.9	18.7930	17.9231	18.8776	22.5481	24.6155	23.9510
No.10	3.1051	17.3298	16.4457	10.1823	18.6101	**19.5081**
No.11	-4.7379	15.7932	16.5010	11.3187	18.4794	**19.2174**
No.12	18.3815	16.0982	14.2814	21.7972	**22.3329**	20.9626
No.13	15.5245	10.6338	12.5319	18.6991	**18.8489**	17.3923
No.14	11.9123	14.3295	17.3919	18.2461	**21.2535**	20.9424
No.15	15.4722	13.6338	14.5055	19.2615	**19.9884**	18.8265
No.16	20.7993	14.6326	22.5571	22.5303	**25.2789**	25.1687
No.17	1.2824	16.0697	20.0880	15.7170	21.8768	**22.3385**
No.18	12.9313	9.5010	14.0142	16.5731	**18.8095**	18.2213
No.19	18.6520	19.2883	23.2020	11.9959	19.1574	**22.3669**
No.20	14.8120	17.6559	22.9835	12.4125	22.6863	**24.0327**
No.21	2.9770	12.6731	21.0901	15.7741	21.9088	**22.4688**
No.22	8.8049	11.1040	17.8656	16.1406	**20.3095**	20.2937
No.23	17.5065	15.6272	16.3651	20.0917	**21.9738**	21.3848
No.24	37.0130	37.9604	41.1950	41.7548	**45.1377**	44.9166
No.25	17.1370	16.2682	25.5285	21.8971	27.0073	**27.3296**
No.26	19.3786	12.9179	19.3614	23.2876	**24.2390**	23.1194
No.27	12.6135	9.2312	13.5227	15.9667	**18.1822**	17.6282
No.28	9.1227	6.8377	12.8870	13.8774	**16.3857**	15.9577
No.29	1.5804	11.0062	12.2872	2.1530	12.6875	**14.0512**
No.30	26.5766	25.9286	28.5463	30.8911	**33.4679**	32.9621

After implementing the three iterations, the comparisons between the polarization coherent synthesis results and the clutter suppression results obtained from **HH** polarization channel are shown in Fig. 27. From the figure, one can see that the clutter suppression performance is significantly improved after polarization coherent synthesis, with an average CNR reduction of 4.86 dB. In the best case, the clutter power can be reduced by 11.63 dB.

Subsequently, the moving target output SCNRs after clutter suppression in each polarization channel and after multiple iterations of polarization coherent synthesis are analyzed. Based on 30 moving targets identified in Fig. 28, the SCNR for each target is calculated for **HH**, **HV**, and **VV** polarization channels after clutter suppression and multiple iterations of polarization coherent synthesis, as shown in Table I. Statistical analysis indicates the significant SCNR differences among different polarization channels. According to Fig. 23(d)–(e), it can be seen that while the **HV** polarization channel exhibits relatively low clutter energy, it also suffers from low target energy, resulting in relatively lower SCNR. In addition, although the co-polarization channels show strong target energy after clutter spatial-domain suppression, the higher residual clutter energy is also accompanied, increasing the false alarm rate.

Comparing the results after polarization coherent synthesis, the output SCNRs of most of moving targets are significantly improved. In addition, the polarization coherent synthesis results based on the targets detection result in the **HH** polarization channel during the first iteration may still contain estimation bias. Multiple iterations allow for the more accurate estimation of the target polarization information. After multiple iterations, as shown in Table I, the target output SCNRs improve markedly and stabilize after the second and third iterations, indicating that three iterations are sufficient for the moving target detection.

C. Moving Target Polarization Detection Performance Analysis

For the detection results obtained from the aforementioned polarization coherent synthesis, a polarization distance detection window is further designed for secondary target detection. Traditional CFAR relies solely on the target amplitude information, which may lead to a high false-alarm rate in the complex clutter environments. By utilizing the polarization

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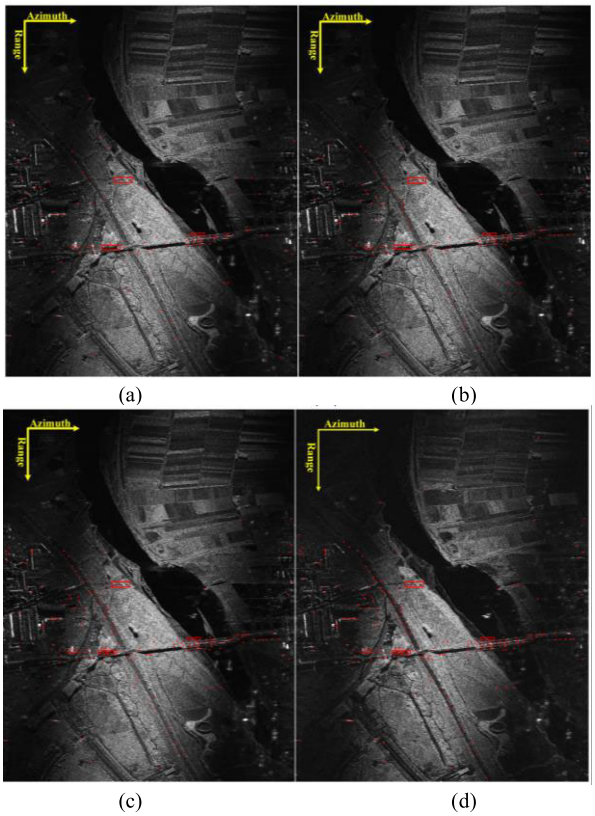


Fig. 26. Target detection results with a CFAR detection threshold of 13 dB, where the red boxes indicate the moving targets that are not detected in the **HH** polarization channel. (a) Target detection results from the clutter spatial-domain suppression image result with the respect to the **HH** polarization channel. (b) Target detection results from the first polarization coherent synthesis iteration. (c) Target detection results from the second polarization coherent synthesis iteration. (d) Target detection results from the third polarization coherent synthesis iterations.

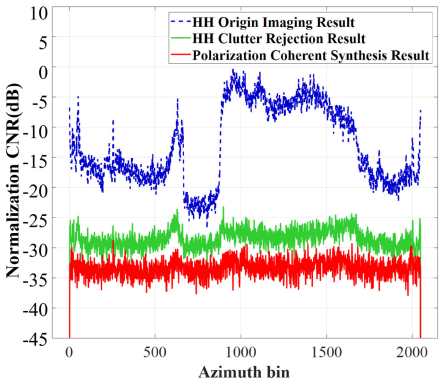


Fig. 27. Comparison of normalized CNR for the original data, the **HH** channel clutter-suppressed result, and the coherently synthesized three-polarization clutter-suppressed result.

differences between the moving targets and residual clutter components, a polarization-distance detection window is designed to further suppress the false alarms.

In the following, the sliding window is applied to the background clutter regions to statistically analyze the distribution of polarization distances, as shown in Fig. 29(a) and (b). The polarization distance data are fit with various

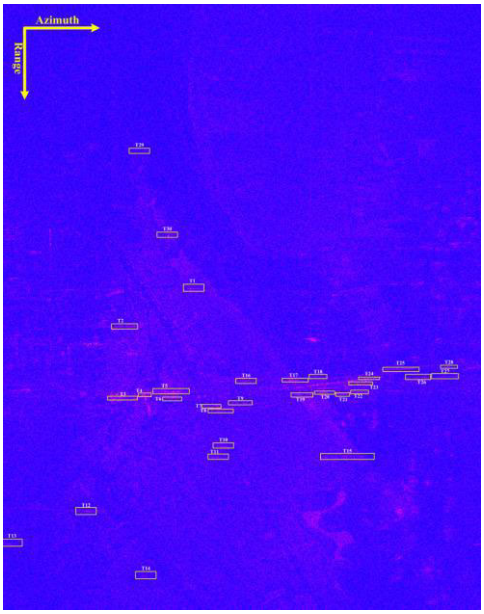


Fig. 28. Moving target detection results after polarization coherent synthesis, with 30 targets, T1 to T30, marked by yellow boxes.

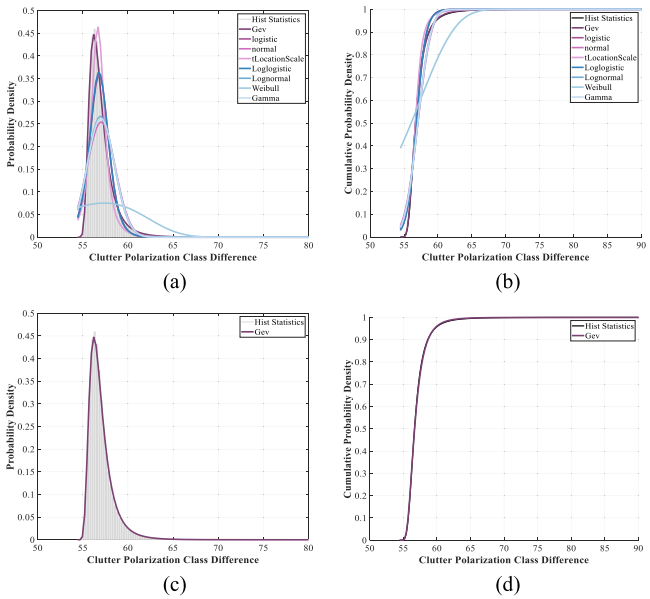


Fig. 29. Clutter polarization distance probability distribution. (a) Statistical distribution of clutter polarization distance and probability density function (pdf) fitting with various distributions. (b) CDF fitting of clutter polarization distance with various distributions. (c) PDF of clutter polarization distance along with its optimal GEV fit. (d) CDF of clutter polarization distance along with its optimal GEV fit.

distributions including generalized extreme value (GEV) [66], logistic [67], normal [68], t location-scale [69], log-logistic [70], log-normal [71], Weibull [72], and Gamma distribution [73]. The cumulative distribution functions (CDFs) of the empirical data distribution and fit distributions are compared using the Kolmogorov–Smirnov (K–S) test [74] to determine the maximum absolute deviation. The results indicate that the GEV distribution provides the best fit to the polarization distance distribution of the clutter background, as shown in

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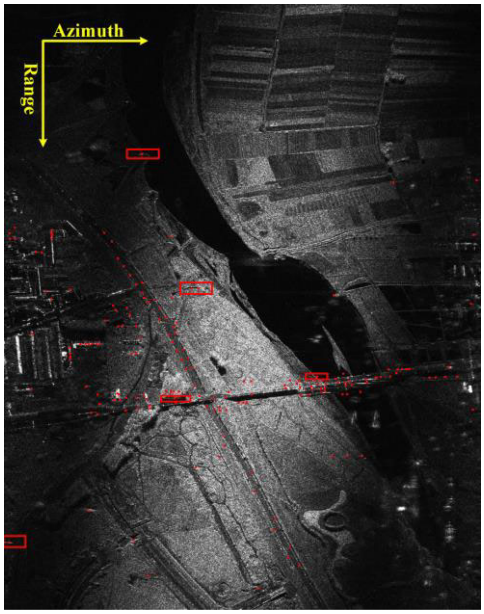


Fig. 30. Detection results of the polarization coherent synthesis by using the polarization-distance detector.

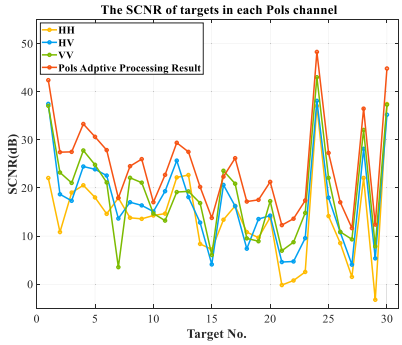


Fig. 31. Target output SCNRs of these 30 moving targets in the **HH**, **HV**, and **VV** channels and after polarization adaptive processing.

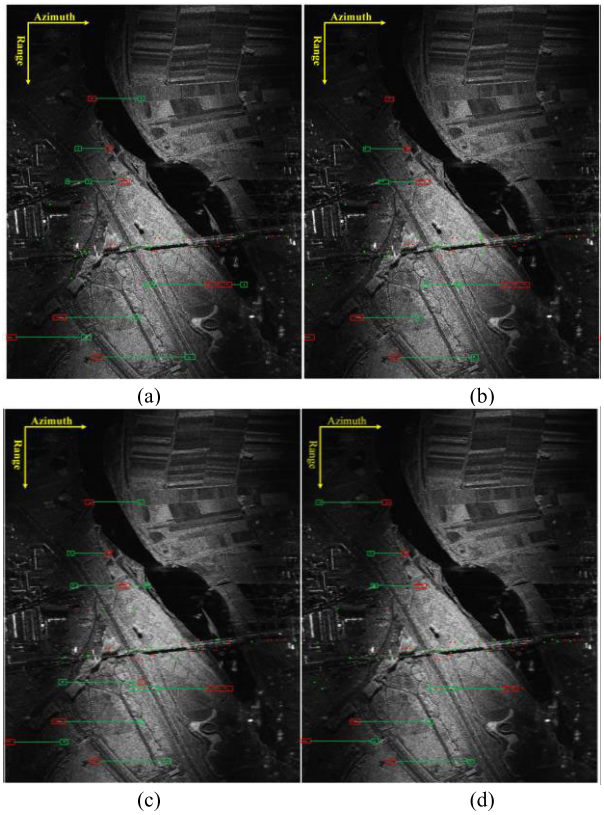


Fig. 32. Target repositioning results, where red markers indicate the detected moving targets and green markers denote the corresponding repositioning results. (a) Repositioning results obtained from the original three **HH** polarization channels data. (b) Repositioning results obtained from the original three **HV** polarization channels data. (c) Repositioning results obtained from the original three **VV** polarization channels data. (d) Repositioning results after polarization synthesis processing.

Fig. 29(c) and (d). According to the fit GEV distribution, when $P_{Gev}(X < x_0) = 99.5\%$, i.e., the polarization distance is less than x_0, the sample can be regarded as clutter with a probability of 99.5%.

The target detection results before and after using the polarization distance window filter are shown in Fig. 30. After applying the polarization distance window, the number of detected targets is reduced from 216 to 176. Compared with the 30 moving targets labeled in Fig. 28, no missed detections occur. This demonstrates that the polarization distance detector proposed in this article can utilize the differences in polarization properties between the moving targets and background clutter components, so as to significantly reduce false alarms.

D. Target Localization Analysis

Finally, the target detection results are further analyzed for repositioning. Specifically, the adaptive matched filter (AMF) method [75] is sequentially applied to estimate the target radial velocity, and target repositioning is then performed based on the estimated target radial velocity. It is important to note that the repositioning operation should be applied to the spatial domains of the three channels in the original imaging data. However, based on the analysis of moving target output SCNRs across different polarization channels, we can know that the target amplitude varies across polarization channels. Considering that the high SCNR can effectively improve the target radial velocity estimation precision, the adaptive processing in the polarization domain is applied using the estimated target polarization steering vector, so as to enhance the target output SCNR and improve the target localization performance. As for these 30 moving targets labeled in Fig. 28, the SCNR values obtained from the **HH**, **HV**, and **VV** polarization data, as well as from polarization adaptive processing data, are shown in Fig. 31. From the figure, it is seen that most of moving targets exhibit a noticeable SCNR improvement after polarization adaptive filtering. Fig. 32 shows the target localization results, where the positioning precision in the **HV** cross-polarization channel is relatively inferior due to the relatively weak target intensities. Although the positioning performance in the **HH** and **VV** polarizations are improved compared with Fig. 32(b), some noticeable positioning deviations are still present. In contrast, the polarization-adaptive processed data exhibits the best positioning quality, indicating that the polarization-adaptive

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processing can significantly improve the target radial velocity estimation as well as repositioning performances.

V. Conclusion

This article discusses the technology of polarized multichannel GMTI based on real-measured data from an airborne N-SAR system. Initially, the echo data is preprocessed for SAR imaging, followed by spatial polarization cascaded clutter suppression. The performance of the proposed polarized multichannel GMTI framework is then validated through the moving target detection and repositioning results. The experimental results demonstrate that: 1) the JPDPCA-based clutter suppression method proposed in this article, assisted by polarization classification and GIP weighting, can effectively improve the clutter rejection and target detection performances, even in the inhomogeneous clutter scene; 2) the iterative estimation of the target polarization steering vector and polarization subspace enables effective polarization-domain coherent synthesis across three polarization channels, significantly reducing the clutter energy while improving the moving target output SCNR; 3) the proposed polarization-distance window detector further reduces false alarms based on the polarization differences between moving targets and background clutter; and 4) polarization adaptive processing of three-channel data, based on the estimated target polarization steering vector, can enhance the target SCNR, significantly improving the target radial velocity estimation and repositioning performances. The reliability and feasibility of the proposed algorithm are validated through the analysis of real-measured SAR data.

Despite the promising detection performance achieved on real-measured data, several limitations of the proposed method should be acknowledged. First, the proposed method relies on the accuracy of polarization classification. When the polarimetric scattering characteristics of different terrain types overlap significantly, classification errors tend to propagate to subsequent covariance matrix estimation and adaptive processing stages, thereby degrading overall performance. Second, although certain parameters in our method (e.g., the CFAR detection threshold and the polarization distance window threshold) are adaptively determined based on statistical models, they may still require readjustment when the scene characteristics undergo drastic changes.

In summary, compared to traditional multichannel SAR-GMTI systems, polarized multichannel SAR-GMTI systems demonstrate superior target detection and localization performances. The proposed cascade processing framework of spatial and polarization domains can achieve the good balance between the computational complexity and processing performance. Future work will focus on achieving joint optimal adaptive processing across spatial and polarization domains with reduced computational complexity. In this regard, knowledge-aided polarization classification methods, or leveraging external geographic information and prior scattering models, represent a promising direction to improve classification accuracy and further strengthen algorithmic robustness in complex scenarios.

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